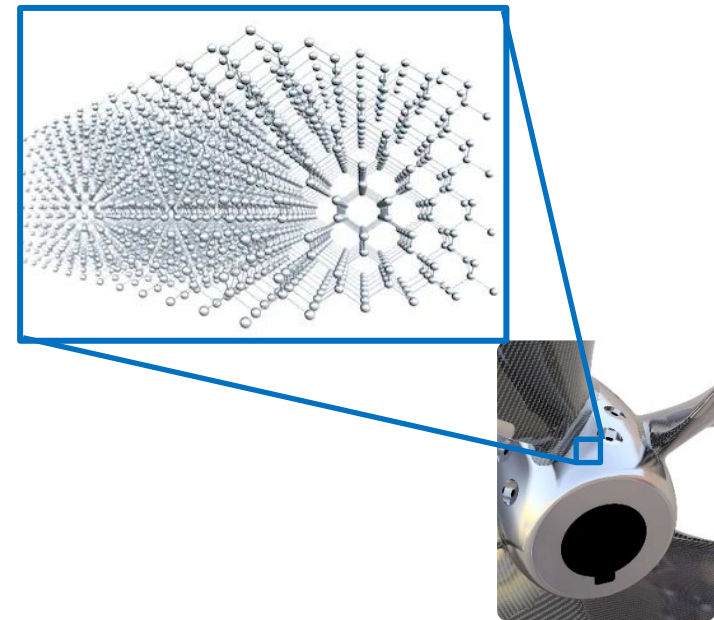


TOPIC 2. STRUCTURE OF MATERIALS I

- **Crystalline systems**
- **Atomic packing**
- **Main metallic structures.**
- **Interstitial sites**
- **Atomic positions.**
- **Directions in the crystalline cells.**
- **Planes in the crystalline cells.**



CRYSTALLINE SYSTEMS

*Why do we have to study
The structure of materials?*

Boring? Well,

Performance of a component = f (properties of the material)!!

Properties of a material = f (**structure**)!!



Wings material?

Ceramic (diamond)?



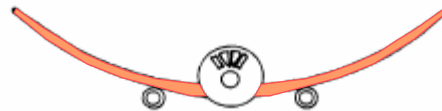
Too brittle ❌

Conner?



Too weak ❌

Polymer (rubber)?



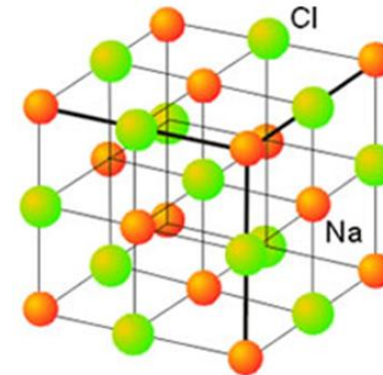
Too flexible ❌

Structure: **Arrangement** of the
components of the material

CRYSTALLINE SYSTEMS

CRYSTALLINE SOLIDS

- Most materials are in crystalline state:
 - ✓ Metals and alloys
 - ✓ Most ceramic materials

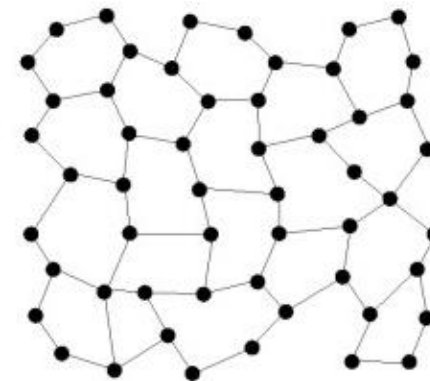


Periodic long-range order of atoms → 3D structure

AMORPHOUS SOLIDS

Amorphous ≡ Non crystalline

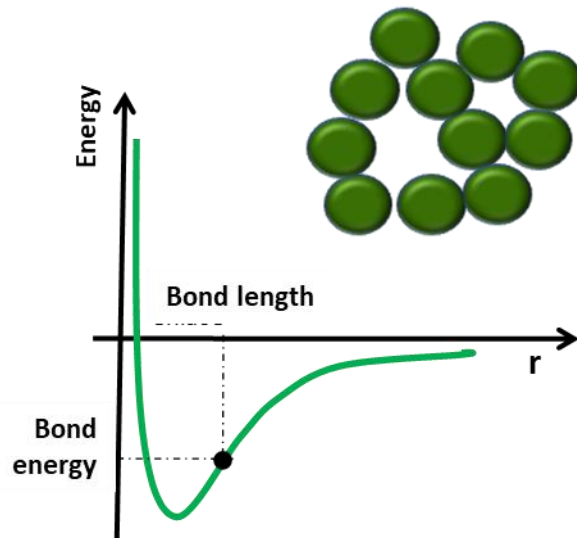
- They do not maintain a periodic ordering
- Solids are amorphous **only if they can't** form crystals for some reason!
 - ✓ Glasses
 - ✓ Some polymers



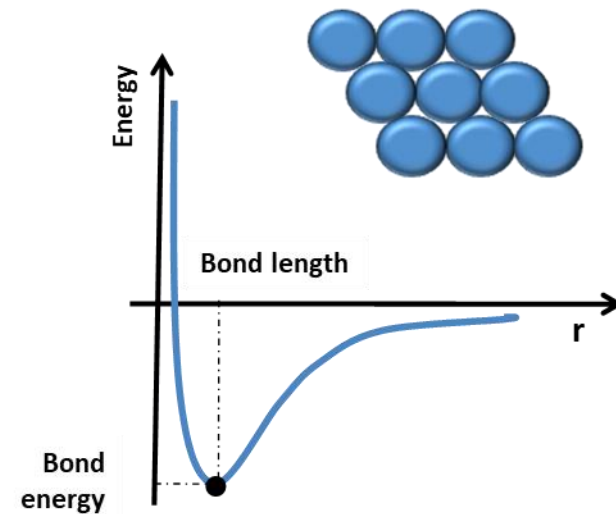
Lack of symmetry and long-range order of atoms

CRYSTALLINE SYSTEMS

- Non dense, random packaging

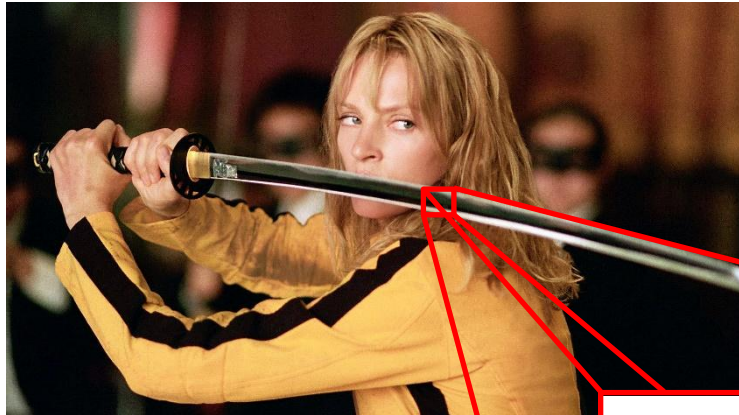


- Dense, ordered packaging



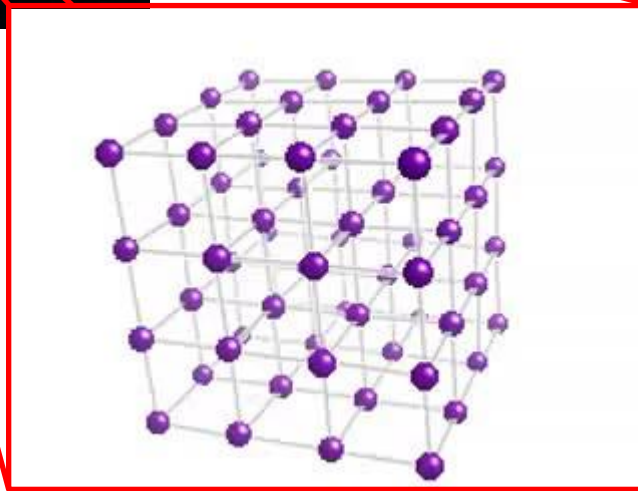
- Atoms tend to assume **positions of minimum energy**
- This results in uniform & repetitive ordering of atoms which is called **crystalline lattice**

CRYSTALLINE SYSTEMS



How are atoms arranged in a “Hattori Hanzo” sword? (Or in a metallic material)

Kill Bill,
2003



Crystalline Lattice

Unit Cell

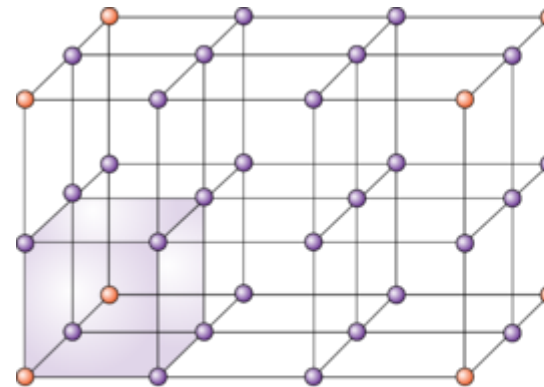
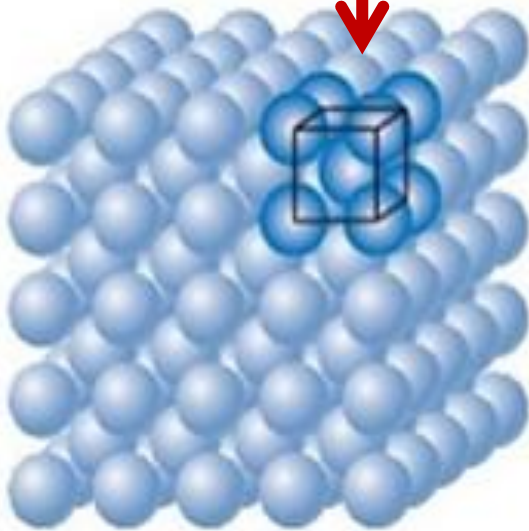
https://www.doitpoms.ac.uk/tlplib/miller_indices/lattice_index.php

CRYSTALLINE SYSTEMS

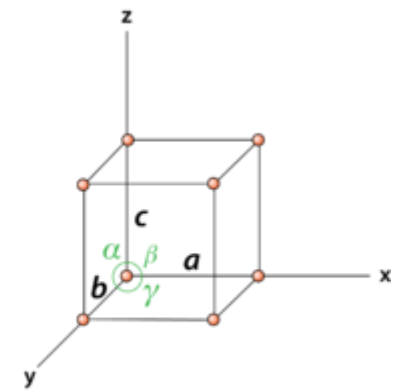
Crystalline lattice: Infinite 3D arrangement of points

Unit Cell: The smallest lattice unit that contains all the elements of symmetry of the crystal.

Parallelepiped capable of reproducing the crystal-“master model”

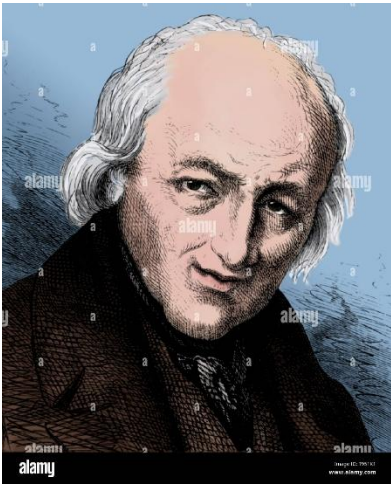


Crystal Lattice



Unit Cell

CRYSTALLINE SYSTEMS



René Just Haüy,
1781



Calcite,
1781



Discovers 2 things:

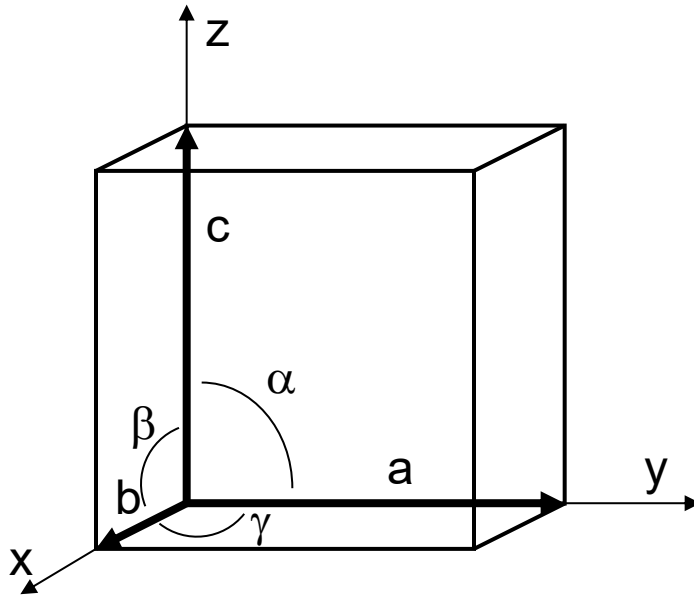
- 1) Calcite is always rhombohedral, no matter how small the shards were
- 2) No voids form whenever he breaks the calcite

Thus, it must be because all calcite small parts are **perfectly packed**

How many mathematically distinct shapes are able to fill 3-D space perfectly with no voids??

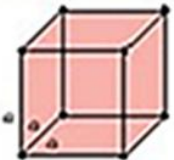
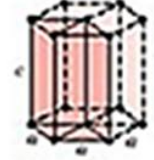
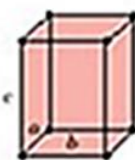
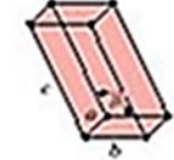
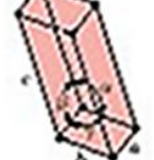
CRYSTALLINE SYSTEMS

UNIT CELL : LATTICE PARAMETERS



- 3 lattice vectors ***a***, ***b*** & ***c*** with common origin
- 6 lattice parameters:
 - ✓ Lengths of edges: *a*, *b*, *c*
 - ✓ Angles between vectors: *α*, *β*, *γ*

CLASIFICATION ACCORDING TO LATTICE PARAMETERS

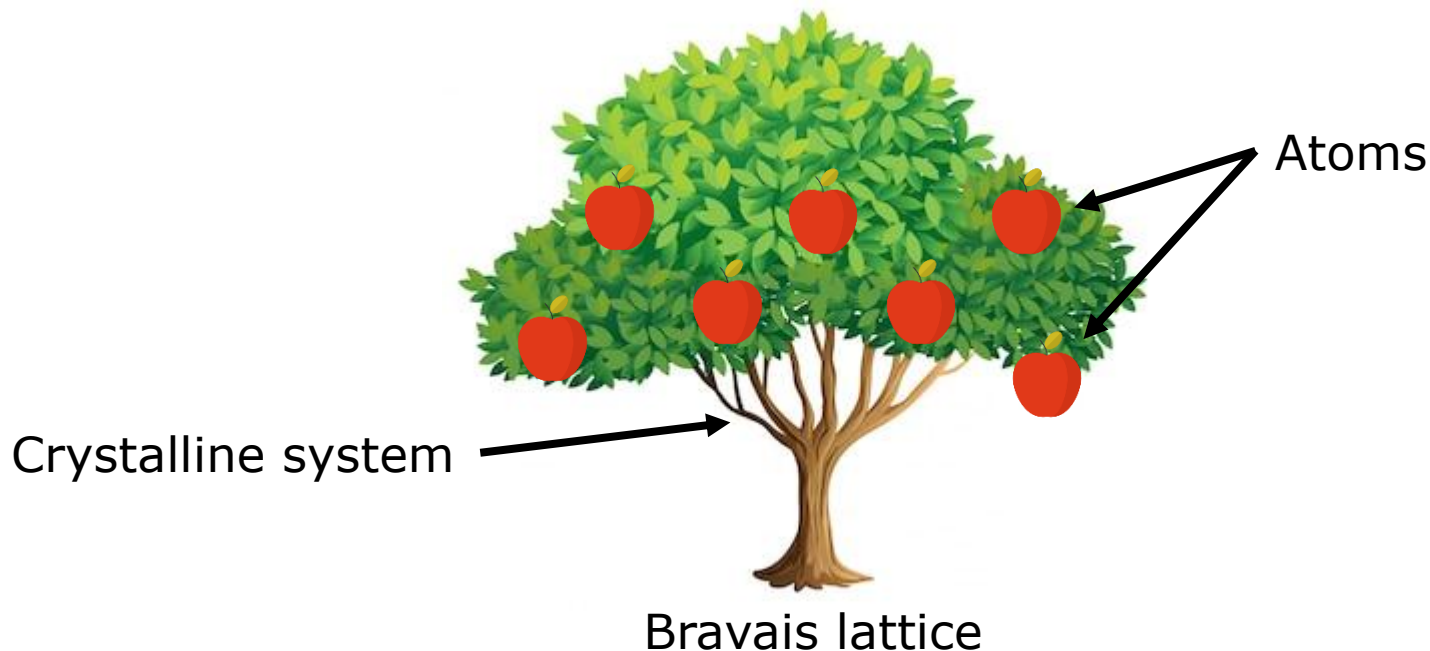
System	Cubic	Hexagonal	Tetragonal	Rhombohedral	Orthorhombic	Monoclinic	Triclinic
Vectors	$a=b=c$	$a=b \neq c$	$a=b \neq c$	$a=b=c$	$a \neq b \neq c$	$a \neq b \neq c$	$a \neq b \neq c$
Angles	$\alpha=\beta=\gamma=90^\circ$	$\alpha=\beta=90^\circ, \gamma=120^\circ$	$\alpha=\beta=\gamma=90^\circ$	$\alpha=\beta=\gamma \neq 90^\circ$	$\alpha=\beta=\gamma$	$\alpha=\gamma=90^\circ, \beta \neq 90^\circ$	$\alpha \neq \beta \neq \gamma$
Unit cell							

CRYSTALLINE SYSTEMS: BRAVAIS LATTICES

Ok, but where are the atoms placed in these 7 systems?

Auguste Bravais (1848) discovered that we can only arrange the atoms in these systems in **14** ways

The positions occupied by the atoms/ions in the crystalline solid are explained through the **BRAVAIS LATTICES**



CRYSTALLINE SYSTEMS: BRAVAIS LATTICES

The crystalline systems define the geometric form of the lattice, but there is also the need to establish the **positions occupied by the atoms/ions/molecules** in the solid → **BRAVAIS LATTICES**

Simple lattices (P)

- One atom per lattice point (e.g., Corners).

Face-centered (F)

- Additional atoms on all faces of the lattice.

Body-centered (I)

- Additional atom in the center of the lattice.

Base-centered (C)

- Two additional atoms on the bases of the lattice.

The combination of the 7 crystalline systems and the 4 unit cells (P, F, C, I) :

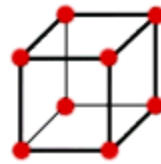
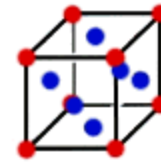
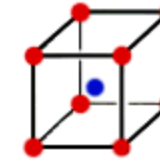
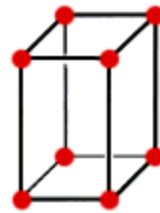
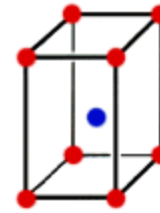
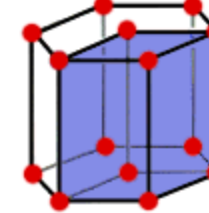
14 fundamental lattices

They are called ***Bravais Lattices***

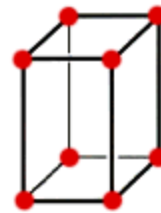
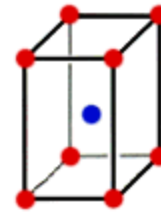
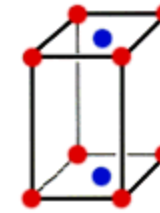
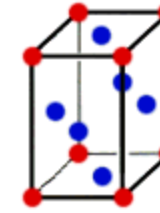
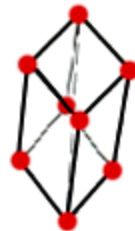
CRYSTALLINE SYSTEMS: BRAVAIS LATTICES



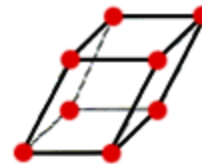
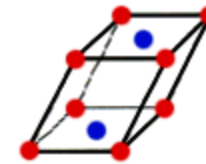
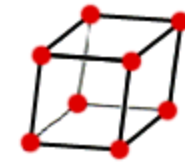
Auguste Bravais

Simple
cubicFace-centered
cubicBody-centered
cubicSimple
tetragonalBody-centered
tetragonal

Hexagonal

Simple
orthorhombicBody-centered
orthorhombicBase-centered
orthorhombicFace-centered
orthorhombic

Rhombohedral

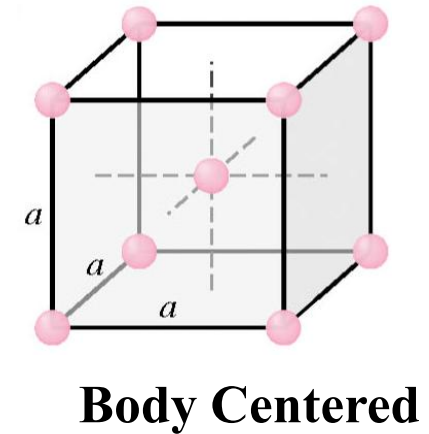
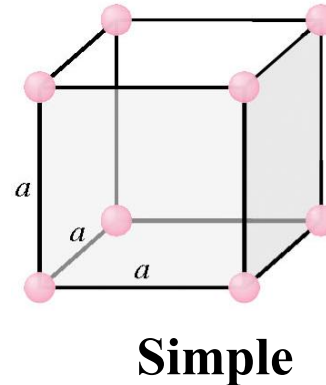
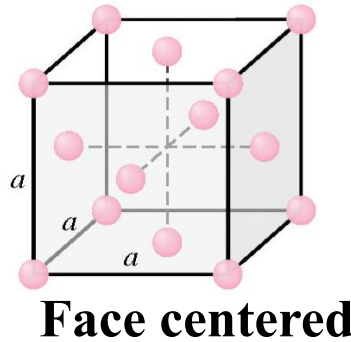
Simple
MonoclinicBase-centered
monoclinic

Triclinic

CRYSTALLINE SYSTEMS: BRAVAIS LATTICES

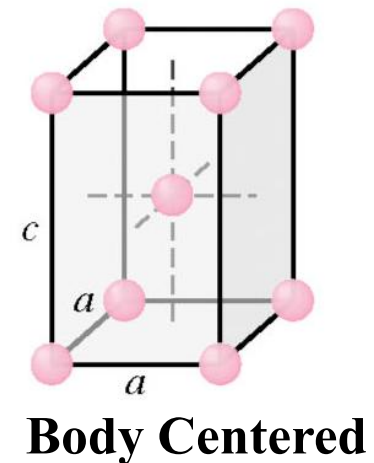
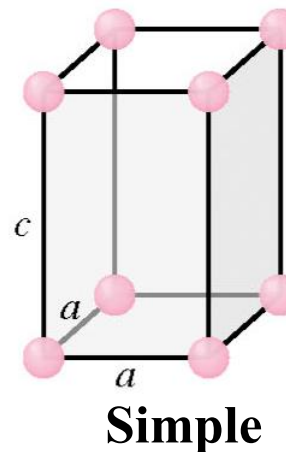
- Cubic Unit Cell**

- $a = b = c$
- $\alpha = \beta = \gamma = 90^\circ$



- Tetragonal**

- $a = b \neq c$
- $\alpha = \beta = \gamma = 90^\circ$

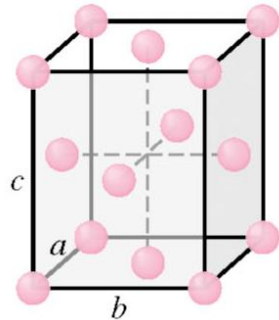


CRYSTALLINE SYSTEMS: BRAVAIS LATTICES

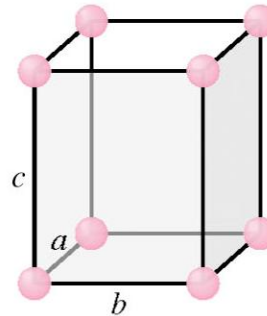
- Orthorhombic**

- $a \neq b \neq c$

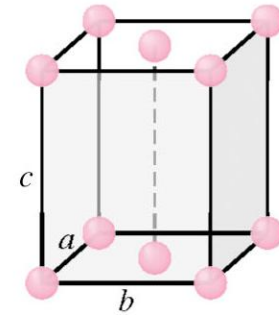
- $\alpha = \beta = \gamma = 90^\circ$



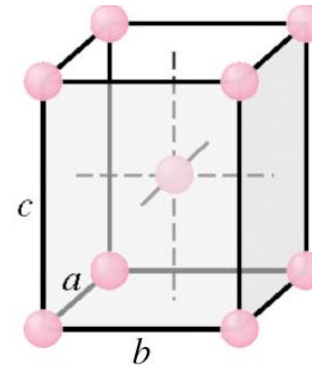
Face Centered



Simple



Base Centered

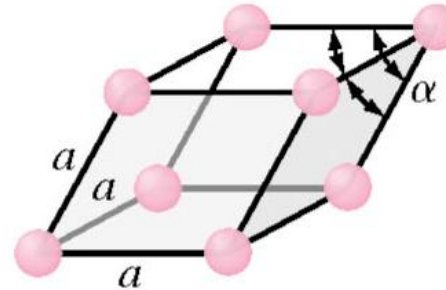


Body Centered

- Rhombohedral**

- $a = b = c$

- $\alpha = \beta = \gamma \neq 90^\circ$

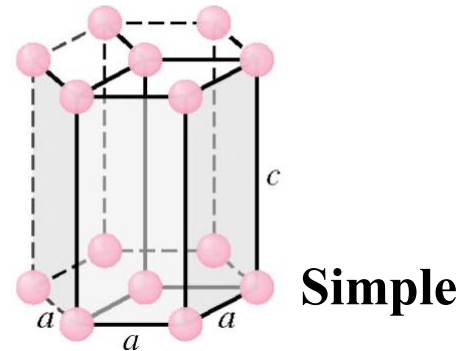


Simple

CRYSTALLINE SYSTEMS: BRAVAIS LATTICES

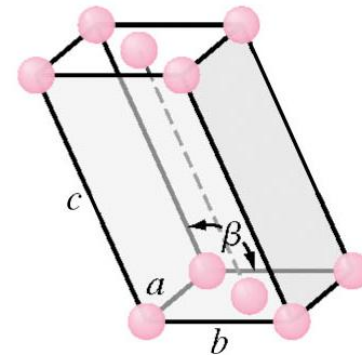
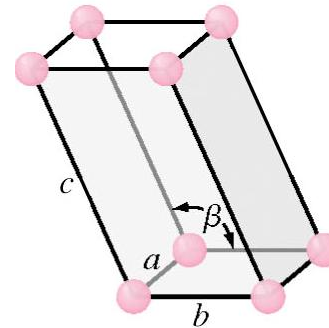
- Hexagonal

- $a = b \neq c$
- $\alpha = \beta = 90^\circ, \gamma = 120^\circ$



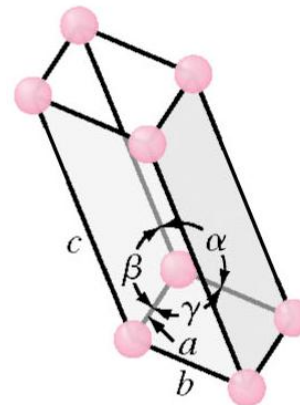
- Monoclinic

- $a \neq b \neq c$
- $\alpha = \gamma = 90^\circ \neq \beta$



- Triclinic

- $a \neq b \neq c$
- $\alpha \neq \beta \neq \gamma \neq 90^\circ$



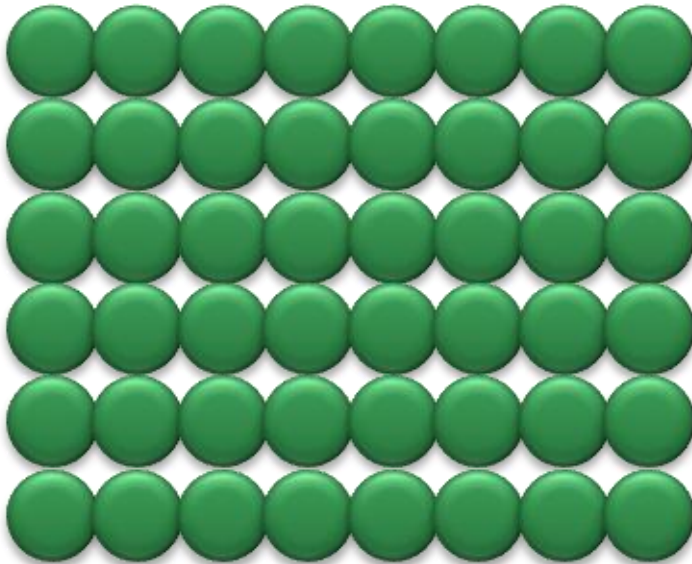


TOPIC 2. STRUCTURE OF MATERIALS I

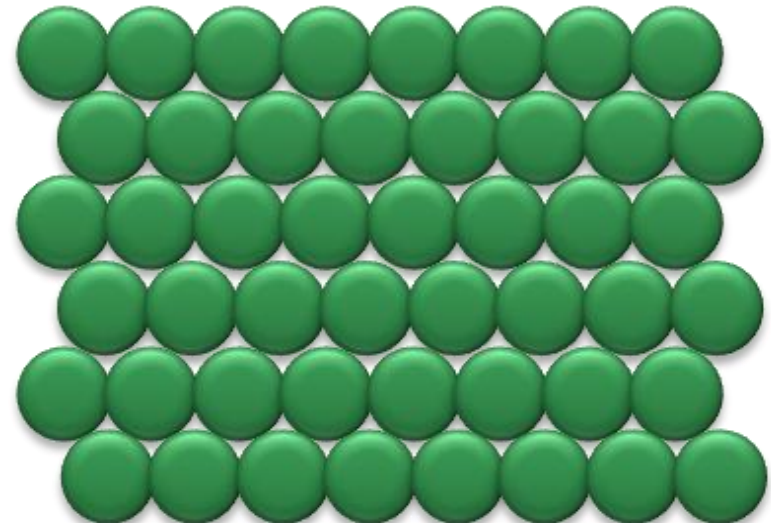
- Crystalline systems.
- **Atomic packing**
- Main metallic structures
- Interstitial sites
- Atomic positions.
- Directions in the crystalline cells.
- Planes in the crystalline cells.

ATOMIC PACKING

Non-compact plane



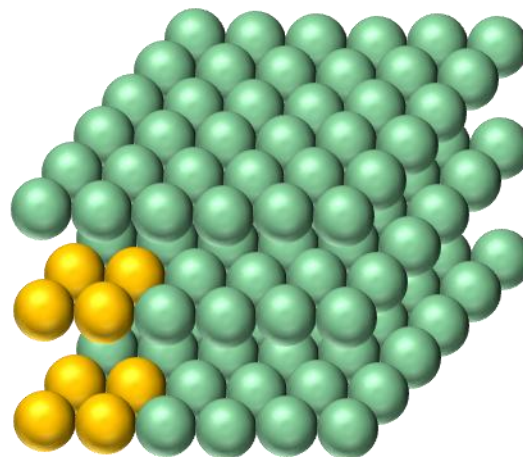
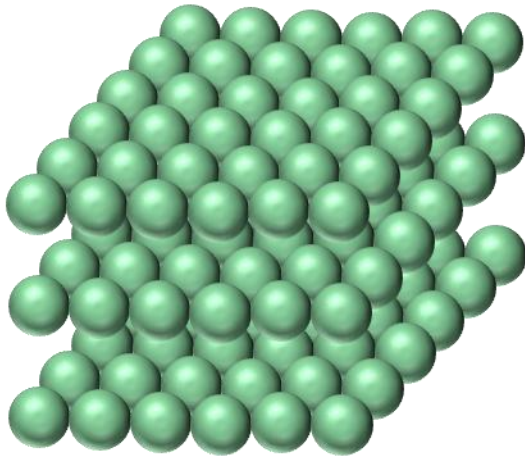
Closed-packed plane



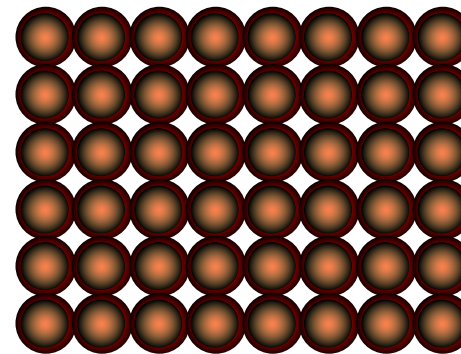
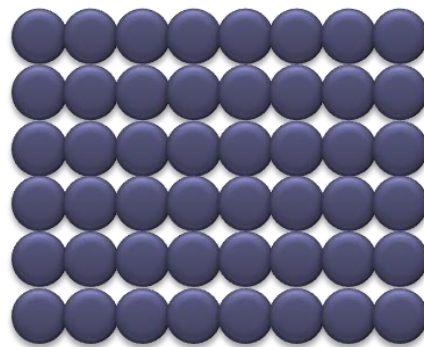
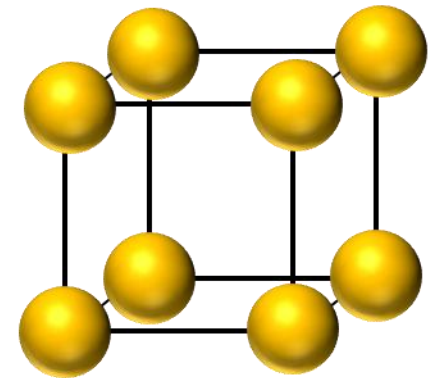
ATOMIC PACKING

Non-compact planes

Non-compact packing



Simple cubic

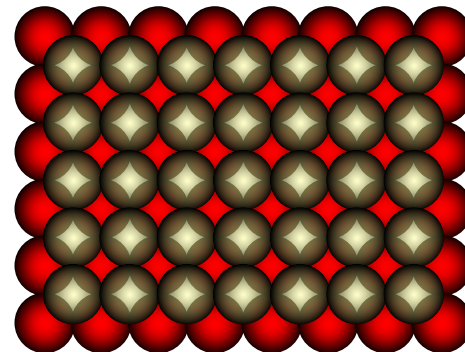
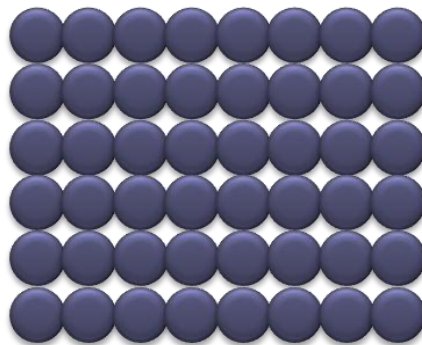
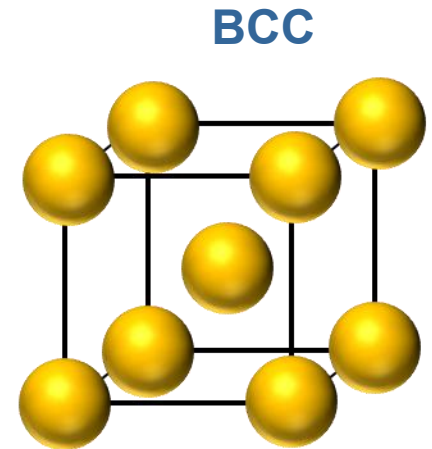
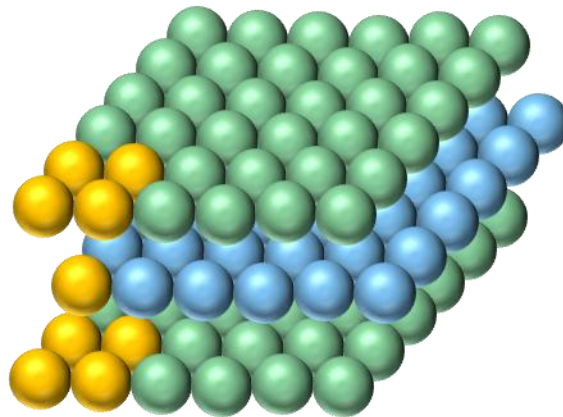
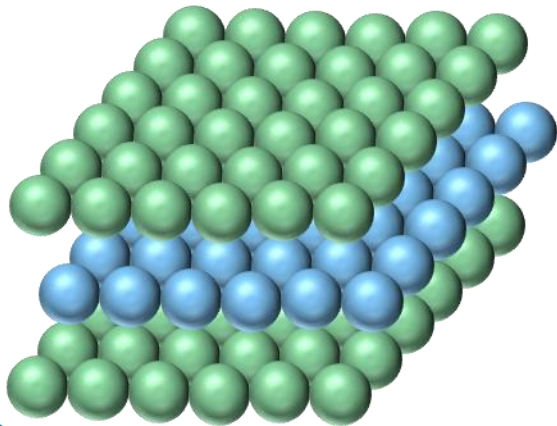


Top view

ATOMIC PACKING

Non-compact planes

Compact packing



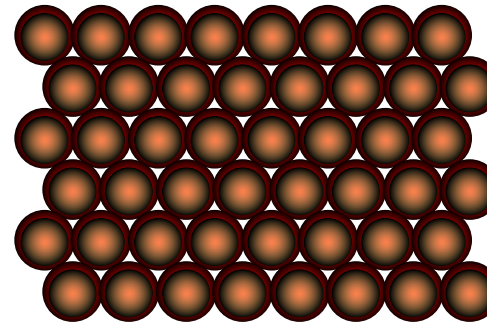
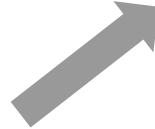
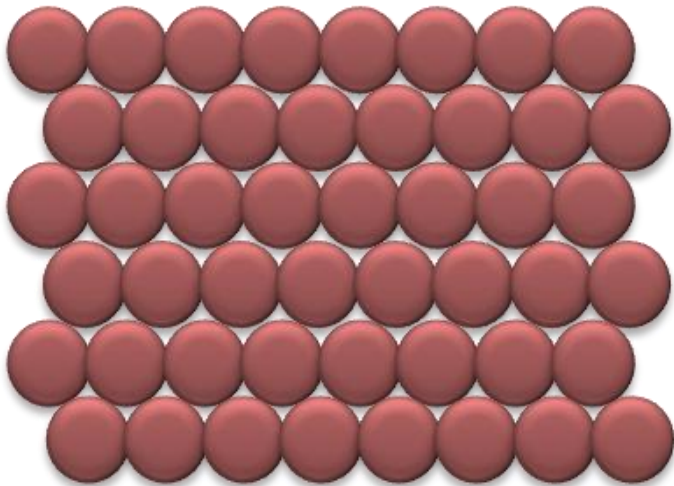
Top view

ATOMIC PACKING

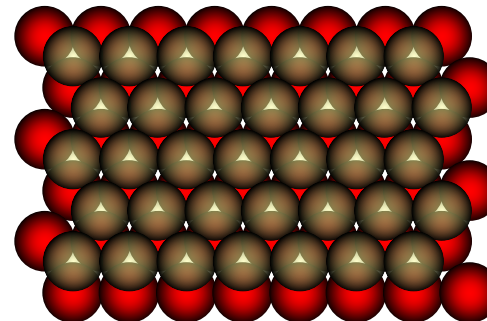
Close-packed planes

PACKING OF LAYERS

Close-packed planes



Non-compact packing



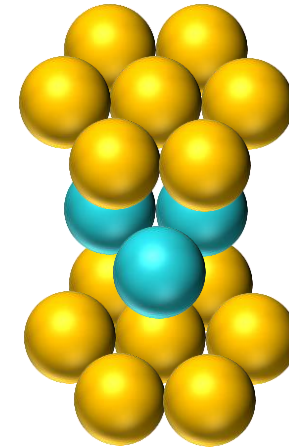
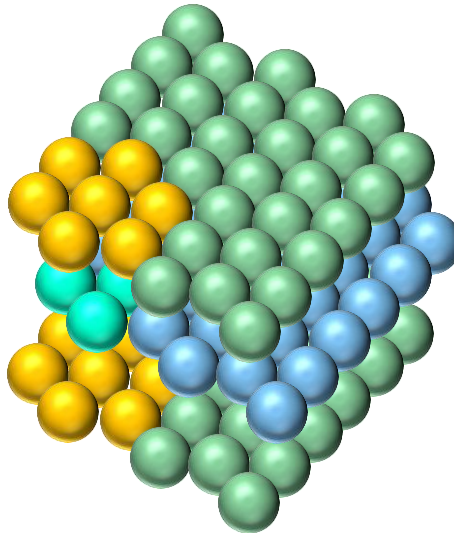
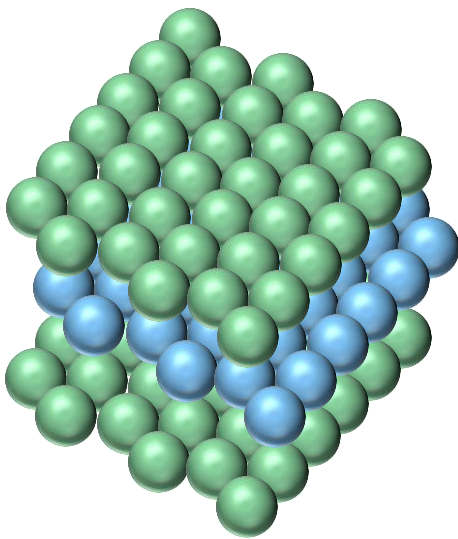
Compact packing

ATOMIC PACKING

Closed-packed planes

Compact packing: stacking sequences

Stacking sequence A B A B A B



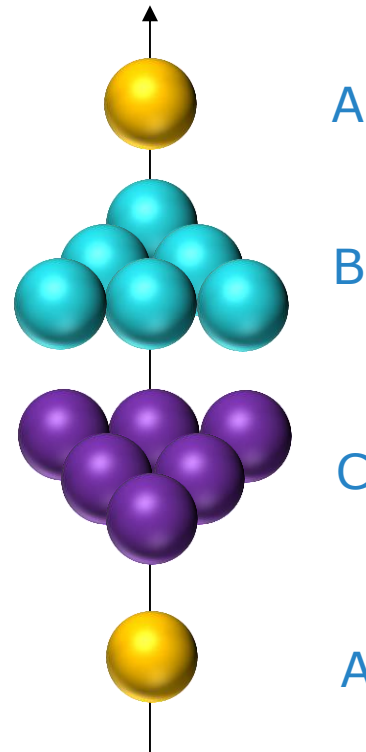
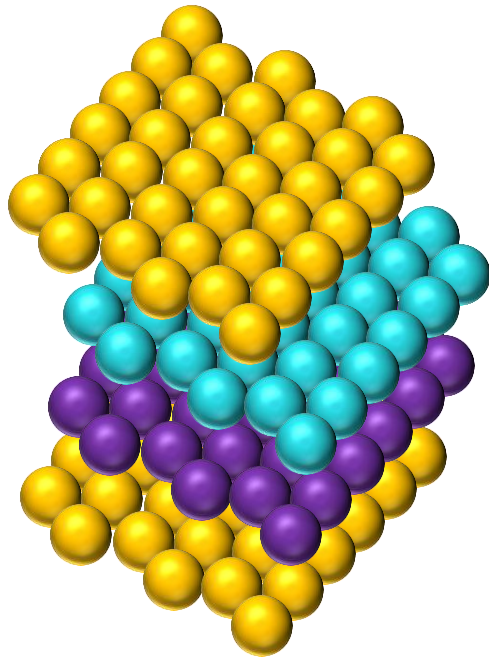
*Hexagonal close
packed structure
(HCP)*

ATOMIC PACKING

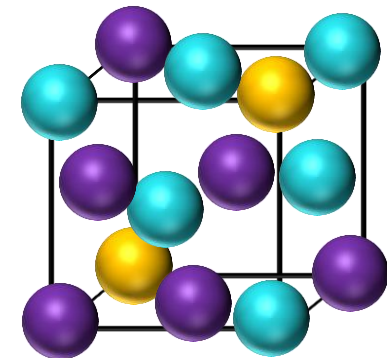
Closed-packed planes

Compact packing: stacking sequences

Stacking sequence **A B A B A B**



A B C A B C

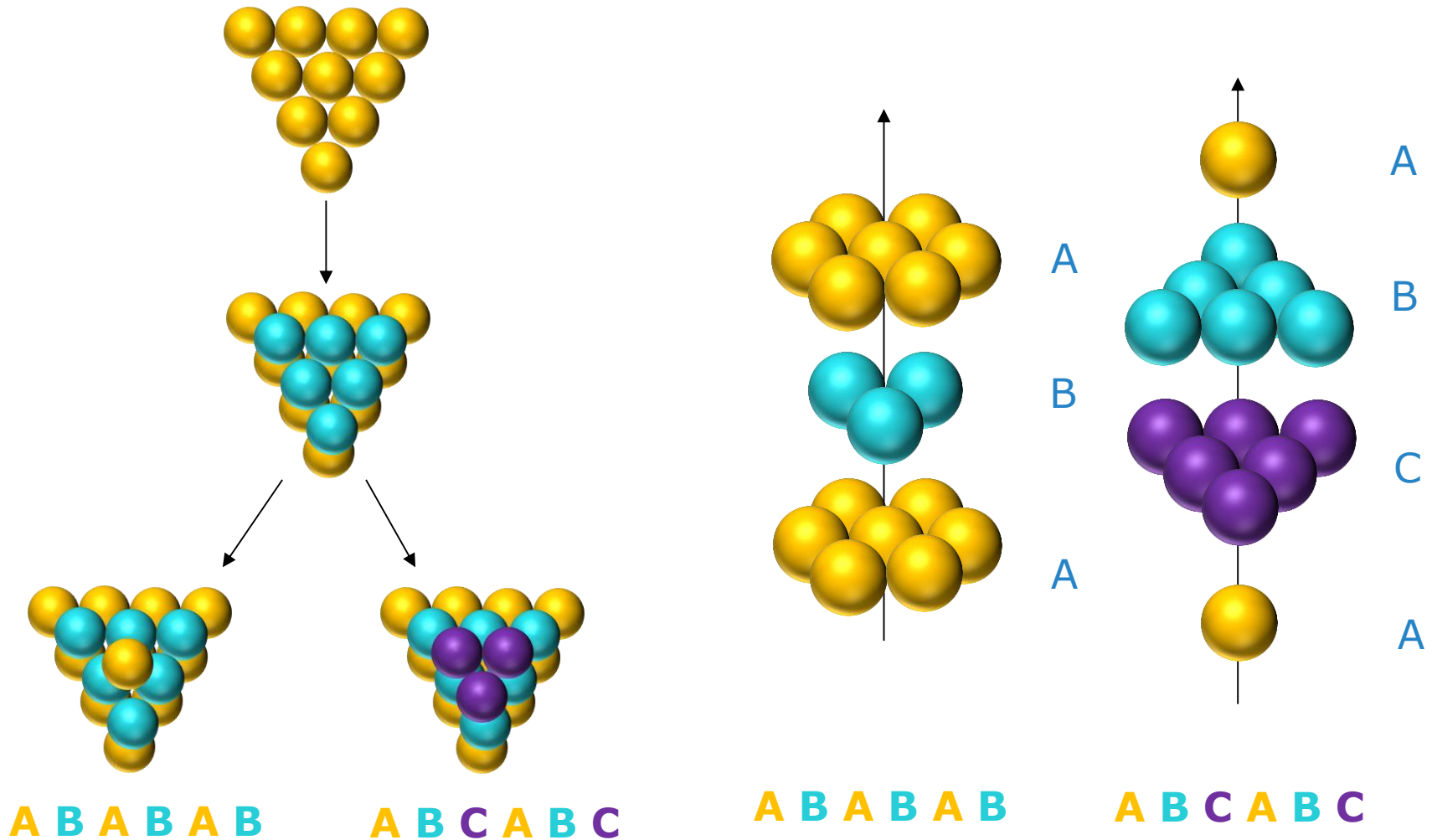


*Face centered cubic
structure
(FCC)*

ATOMIC PACKING

Closed-packed planes

Compact packing: stacking sequences





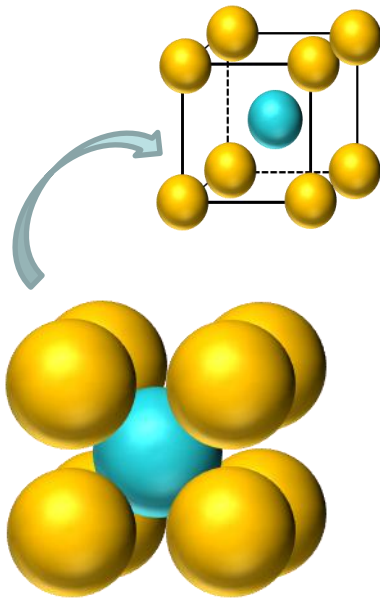
TOPIC 2. STRUCTURE OF MATERIALS I

- Crystalline systems.
- Atomic packing
- **Main metallic structures.**
- Interstitial sites
- Atomic positions.
- Directions in the crystalline cells.
- Planes in the crystalline cells.

MAIN METALLIC STRUCTURES

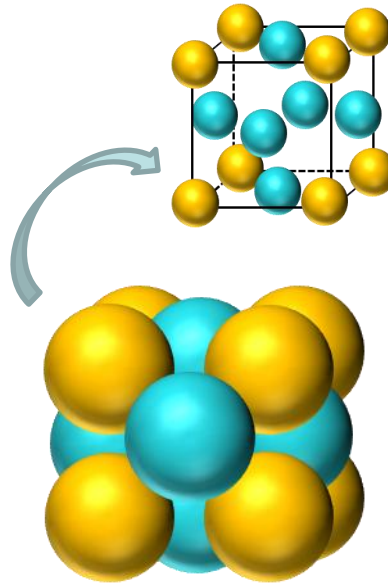
90% of the metals crystallize in 3 densely packed (compact) structures as they :

- Are normally composed by atoms of a single element → same atomic radius.
- The metallic bond is non directional.
- Smaller distances between atoms → higher lattice energies



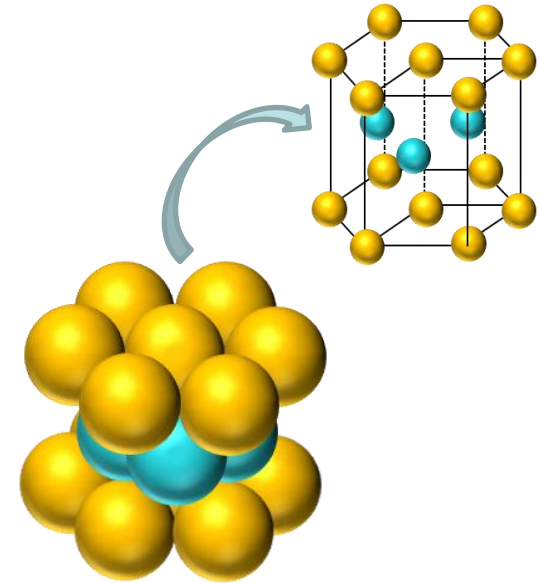
Body-centered cubic
BCC

Fe, Cr, Mo, V, W, K



Face-centered cubic
FCC

Al, Cu, Pb, Ni, Pt, Ag

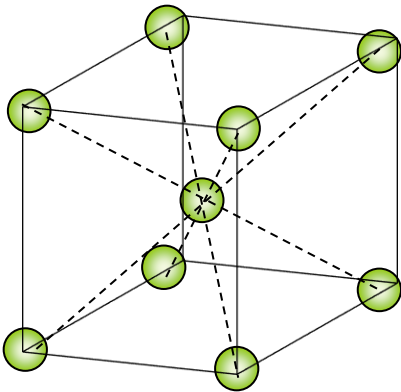
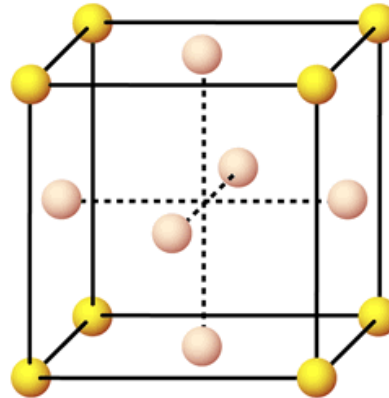
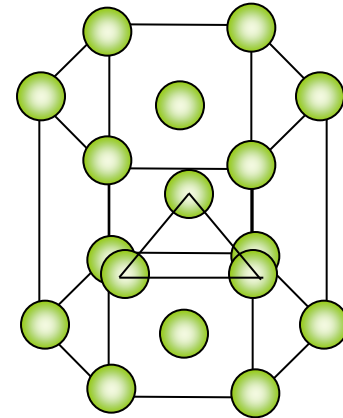


Hexagonal close packed
HCP

Ti, Mg, Cd, Co, Zn

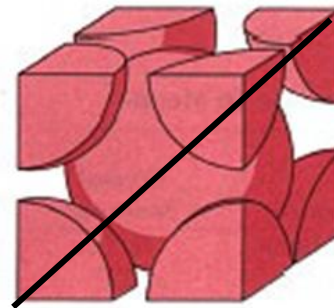
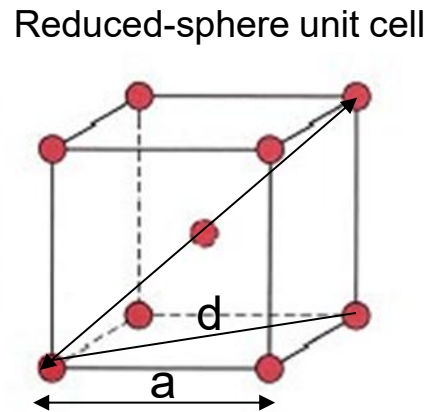
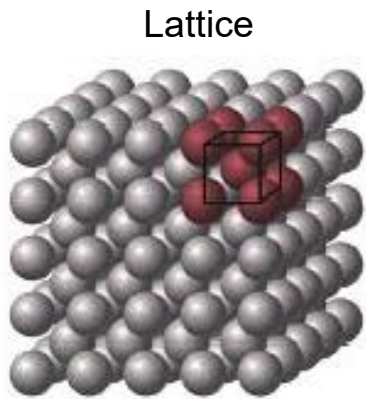
MAIN METALLIC STRUCTURES

- Body- Centered Cubic (BCC)
- Face-Centered Cubic (FCC)
- Hexagonal Close-Packed (HCP)

**BCC****FCC****HCP**

Atomic Arrangement $\Rightarrow E_{\text{minimum}}$

BODY-CENTERED CUBIC STRUCTURE (BCC)



$$\begin{aligned} d &= 2a^2 \\ D^2 &= a^2 + d^2 = 3a^2 \end{aligned} \quad \left. \vphantom{\begin{aligned} d &= 2a^2 \\ D^2 &= a^2 + d^2 = 3a^2 \end{aligned}} \right\} a = \frac{4R}{\sqrt{3}}$$

$$4R = \sqrt{3}a$$

• Number of atoms per cell:

$$n_{atoms} = 1 + 8 \cdot \frac{1}{8} = 2$$

• Coordination number: 8

$$\text{Atomic Packing Factor} = \frac{\text{Volumen of atoms in a unit cell}}{\text{Total unit cell volume}}$$

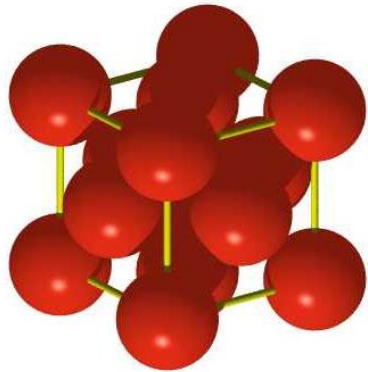
$$V_{atoms} = 2 \cdot \left[\frac{4}{3} \pi \cdot R^3 \right]$$

$$V_{cell} = \left[\frac{4R}{\sqrt{3}} \right]^3$$

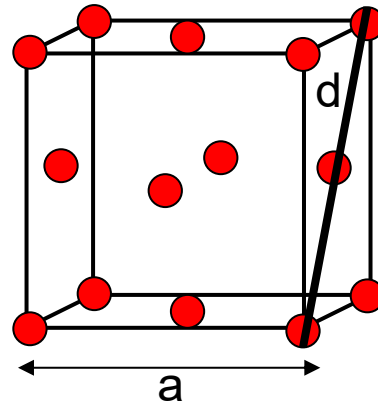
$$APF = \frac{V_{atoms}}{V_{cell}} = \frac{\left[\frac{4}{3} \pi \cdot R^3 \right] \cdot 2}{\left[\frac{4R}{\sqrt{3}} \right]^3} = \frac{\sqrt{3}}{8} \pi = 0.68$$

68 % of the volume of the cell is occupied

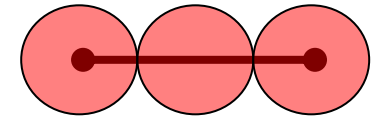
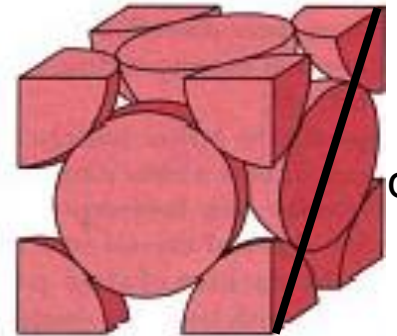
FACE-CENTERED CUBIC STRUCTURE (FCC)



Reduced-sphere unit cell



Hard-sphere unit cell



$$a^2 + a^2 = d^2 \quad d = 4R$$

$$4R = \sqrt{2}a \quad a = 2R\sqrt{2}$$

• **Number of atoms per cell:**

$$n_{atoms} = 8 \cdot \frac{1}{8} + 6 \cdot \frac{1}{2} = 4at$$

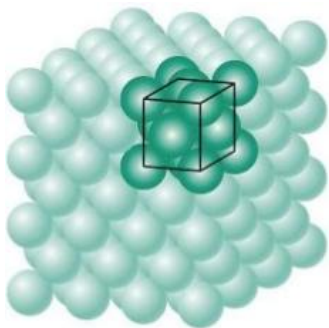
• **Coordination number: 12**

$$\text{Atomic Packing Factor (APF)} = \frac{\text{Volumen of atoms in a unit cell}}{\text{Total unit cell volume}}$$

$$V_{cell} = [2\sqrt{2}R]^3$$

$$V_{atoms} = 4 \cdot \left[\frac{4}{3} \pi \cdot R^3 \right]$$

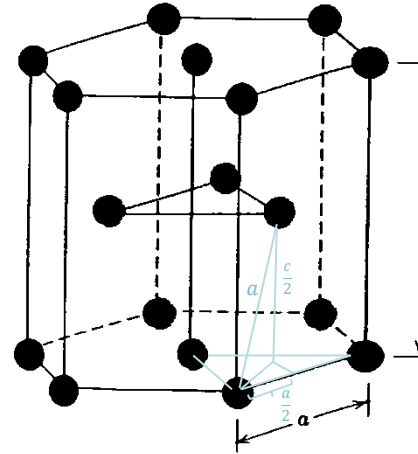
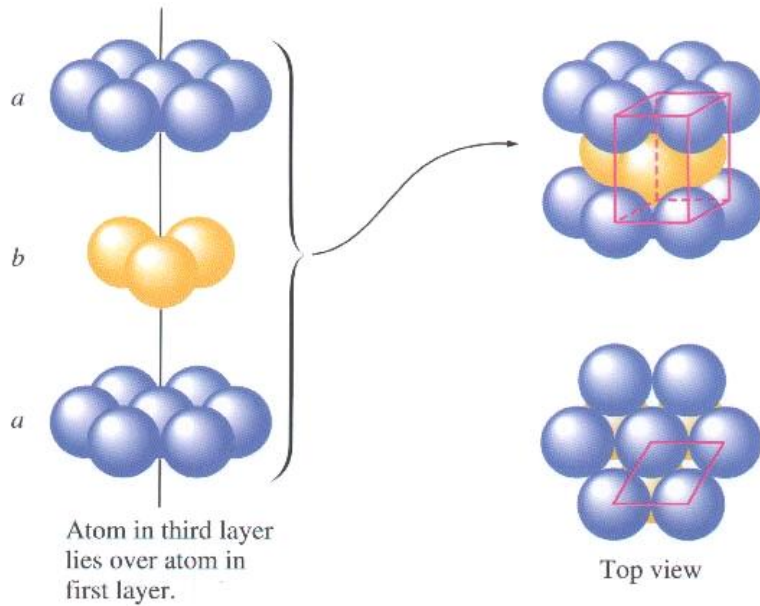
Lattice



$$APF = \frac{V_{atoms}}{V_{cell}} = \frac{\left[\frac{4}{3} \pi \cdot R^3 \right] \cdot 4}{[2\sqrt{2}R]^3} = \frac{\pi}{3\sqrt{2}} = 0.74$$

74 % of the volume of the cell is occupied

HEXAGONAL CLOSED- PACKED STRUCTURE (HCP)



HCP

$$V_{cell} = B \cdot h$$

$$B = 3 \cdot \frac{\sqrt{3}}{2} \cdot a^2$$

$$c = \frac{2 \cdot \sqrt{2}}{\sqrt{3}} a \quad 2R=a$$

Atomic Packing Factor (APF):

$$APF = \frac{V_{atoms}}{V_{cell}}$$

• **Number of atoms per cell:**

$$n_{atoms} = \left[12 \cdot \frac{1}{6} \right] + \left[2 \cdot \frac{1}{2} \right] + 3 = 6at$$

• **Coordination number: 12**

$$V_{atoms} = 6 \cdot \left[\frac{4}{3} \pi \cdot R^3 \right]$$

$$V_{cell} = 24 \cdot \sqrt{2} R^3$$

$$APF = \frac{V_{atoms}}{V_{cell}} = \frac{\left[\frac{4}{3} \pi R^3 \right] \cdot 6}{24 \cdot R^3 \sqrt{2}} = 0.74$$

74 % of the volume of the cell is occupied



TOPIC 2. STRUCTURE OF MATERIALS I

- Crystalline systems.
- Atomic packing
- Main metallic structures.
- **Interstitial sites**
- Atomic positions.
- Directions in the crystalline cells.
- Planes in the crystalline cells.

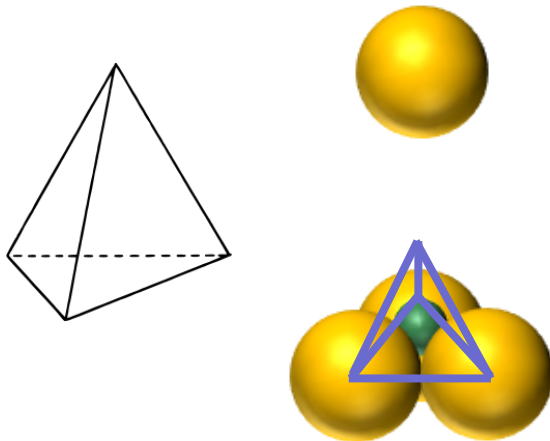
INTERSTITIAL SITES

Interstitial site: a position between the regular positions in a lattice that can be occupied by **other atoms or ions**.

There are 2 types of interstitials sites:

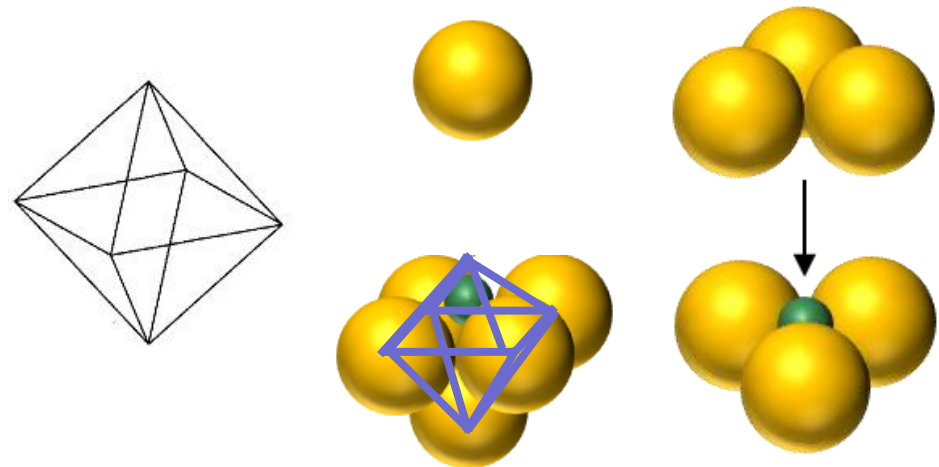
TETRAHEDRAL SITE

(Coordinated by 4 lattice atoms)



OCTAHEDRAL SITE

(Coordinated by 6 lattice atoms)



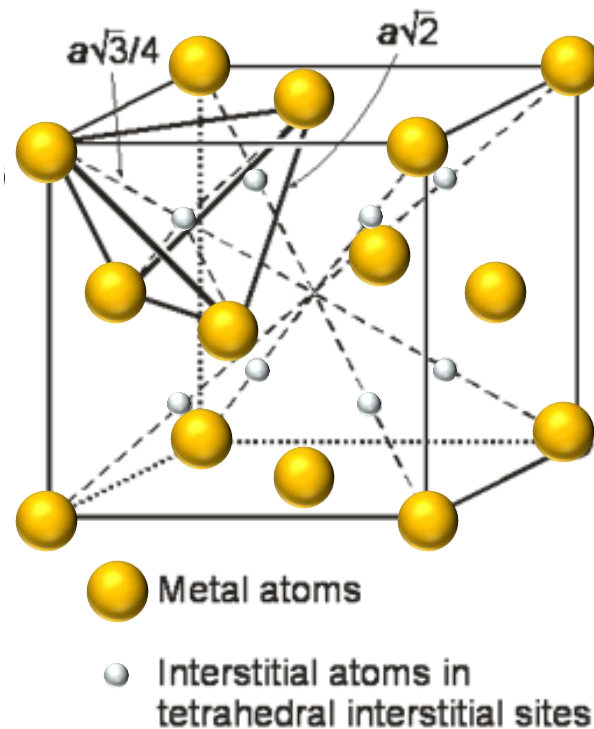
COMPACT structures

- N° of tetrahedral sites = $2n$
- N° of octahedral sites = n
- n = N° atoms in the cell

INTERSTITIAL SITES : IN FCC STRUCTURE

FCC Structure

TETRAHEDRAL SITE

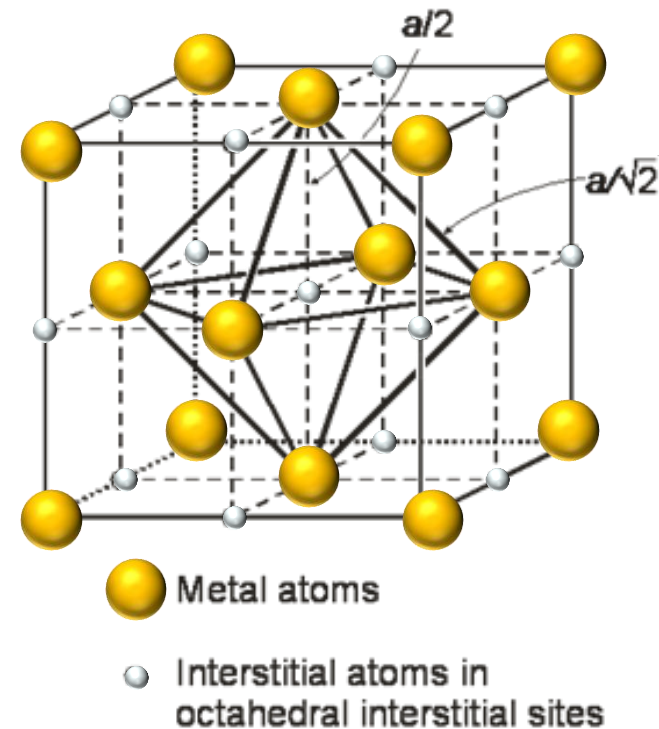


Number of tetrahedral sites in unit cell:

inner cell: $8 \times 1 = 8$

Total: 8

OCTAHEDRAL SITE



Number of octahedral sites in unit cell:

center: $1 \times 1 = 1$

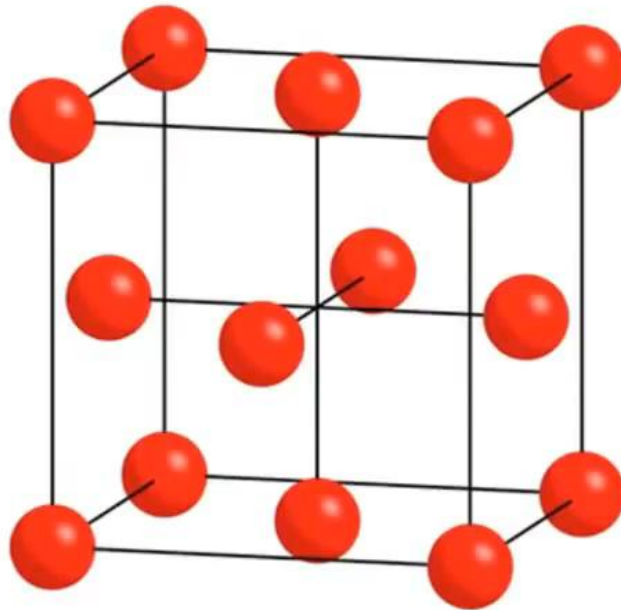
edges: $12 \times \frac{1}{4} = 3$

Total: 4

EASY VISUALIZATION OF INTERSTICES

https://www.youtube.com/watch?v=4AW9PFWaMcQ&ab_channel=OxfordAcademic%28OxfordUniversityPress%29

Cubic close-packed structure



Number of atoms in unit cell

$$\text{vertices: } 8 \times \frac{1}{8} = 1$$

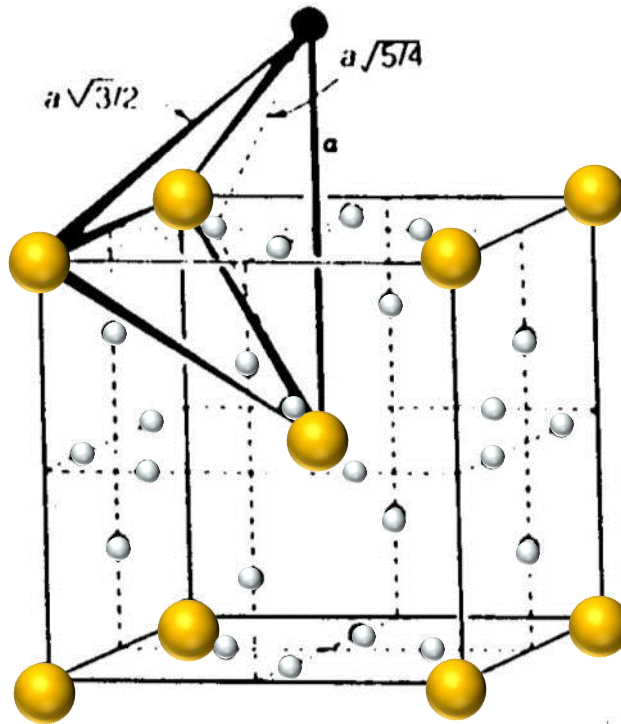
$$\text{faces: } 6 \times \frac{1}{2} = 3$$

$$\text{TOTAL} \quad \quad = 4$$

INTERSTITIAL SITES : IN BCC STRUCTURE

BCC Structure

TETRAHEDRAL SITES

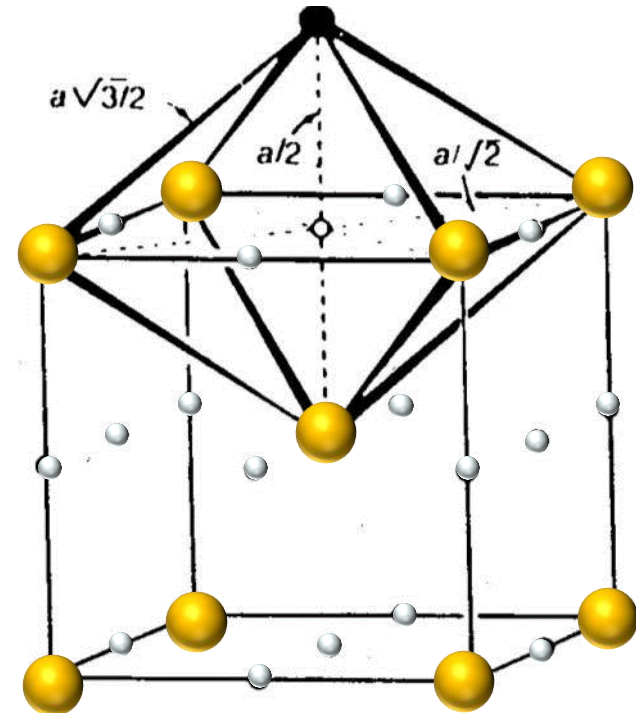


Number of tetrahedral sites in unit cell:

$$\text{faces: } 6 \times 4 \times \frac{1}{2} = 12$$

Total: 12

OCTAHEDRAL SITES



Number of octahedral sites in unit cell:

$$\text{center: } 6 \times \frac{1}{2} = 3$$

$$\text{edges: } 12 \times \frac{1}{4} = 3$$

Total: 6

EASY VISUALIZATION OF INTERSTICES

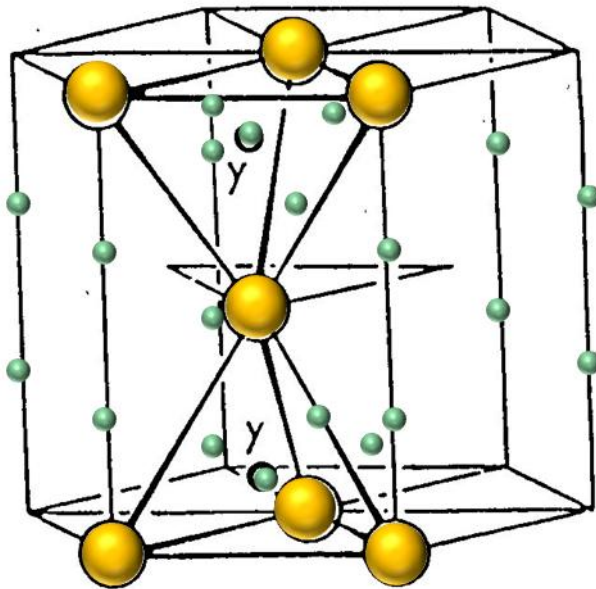
https://www.youtube.com/watch?v=GCA5cY5I_kQ&ab_channel=LearnWithManim

Tetrahedral Voids in BCC Structure

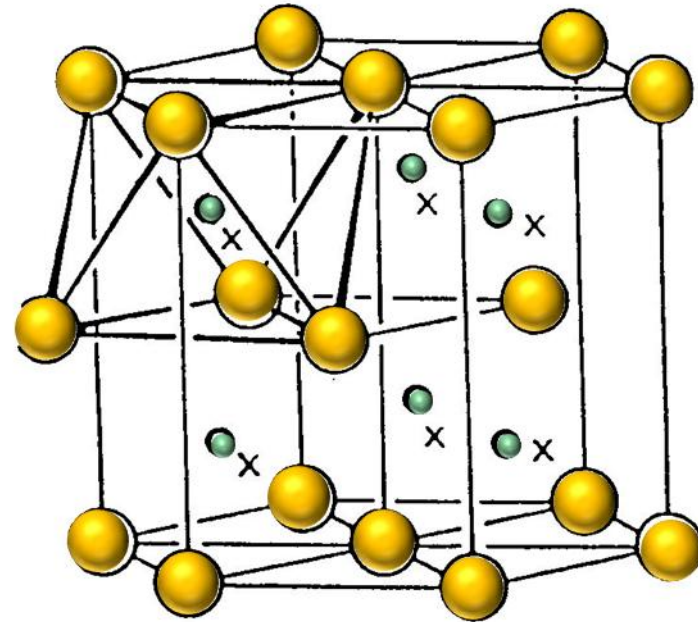
INTERSTITIAL SITES : IN HCP STRUCTURE

HCP Structure

Tetrahedral



Octahedral



Structure

Number of
atomsNumber of
tetrahedral
sitesNumber of
octahedral
sites

FCC

4

8

4

HCP

6

12

6



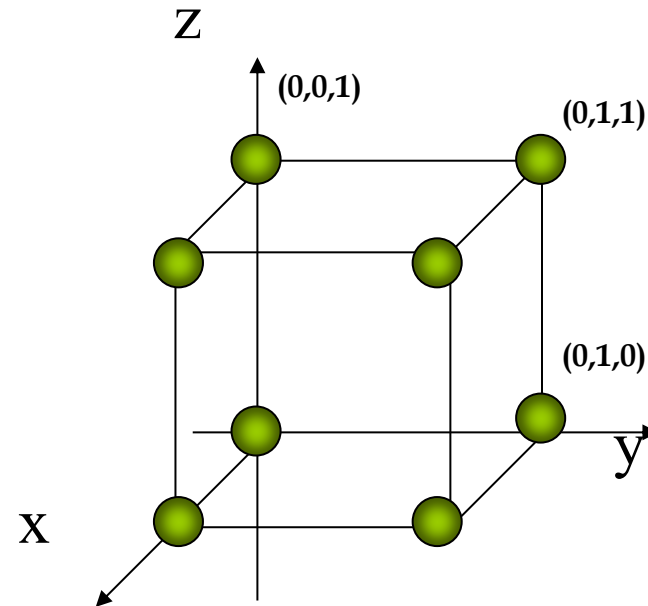
TOPIC 2. STRUCTURE OF MATERIALS I

- Crystalline systems and some definitions.
- Atomic Packing
- Important metallic structures.
- Interstitial sites
- **Atomic positions.**
- Directions in the crystalline cells.
- Planes in the crystalline cells.

ATOMIC POSITIONS IN THE CUBIC CELLS

Atomic positions

- In **orthogonal** lattices ($a=b=c$, $\alpha=\beta=\gamma=90^\circ$) a set of lattice vectors ***a***, ***b*** & ***c*** define **uniquely** a point in space and can be used to define **positions of atoms in the unit cell**.
- Cubic lattice is an orthogonal lattice with $a=b=c$
- Values ranging from 0 to 1
- Atomic position: (***a***, ***b***, ***c***)





TOPIC 2. STRUCTURE OF MATERIALS I

- Crystalline systems and some definitions.
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- Planes in the crystalline cells.

DIRECTIONS IN THE CUBIC CELL

Atomic directions in cubic lattices

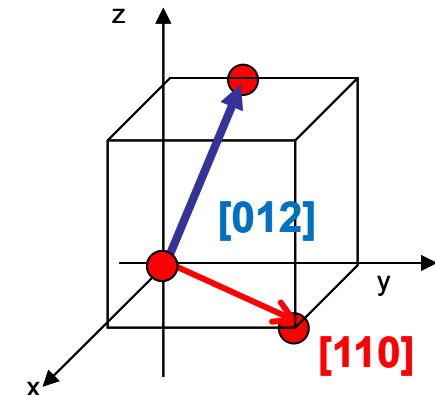
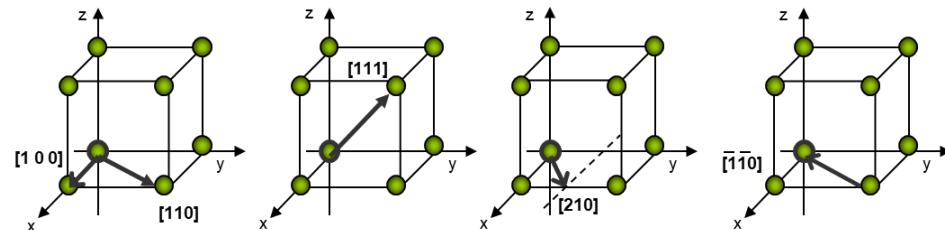
“crystallographic direction indices are the vector components of the directions resolved along each of the coordinate axes and reduced at the smallest integrals”

- **Crystallographic direction:** a vector that connects two points in the lattice
- Atomic direction: $[u \ v \ w]$

Drawing directions from indices

Example 1: $[012]$

1. Select origin **O** at (0,0,0) OR choose a new **O** and label axes x' y' z' (only in case of negative indices)
2. Reduce indices to maximum value of 1 to keep directions within the unit cell: $\frac{1}{2}[0 \ 1 \ 2] = [0, \frac{1}{2}, 1]$
3. Locate from the origin at (0,0,0) a position in the unit cell at $x=0 \ y=\frac{1}{2} \ z=1$
4. Draw direction from O to arrow tip at $0, \frac{1}{2}, 1$

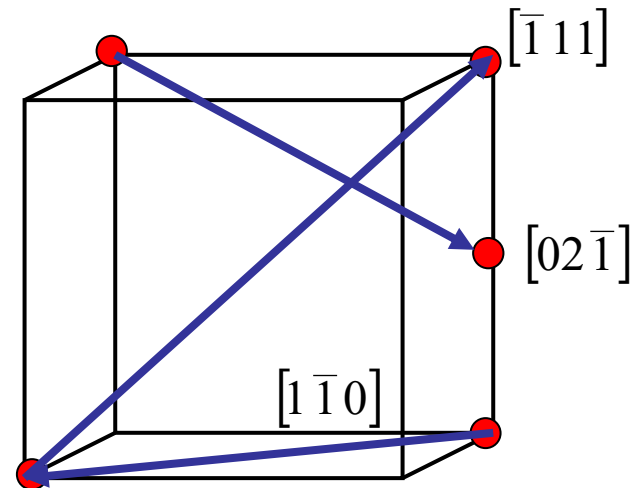
Example 2: $[110]$ 

DIRECTIONS IN THE CUBIC CELL

Directions with negative indices

- They are represented by a bar over the appropriate index: $[\bar{u}vw]$
- If there is a negative index, the new origin O' should have a 1 in the same position as the negative indices and a 0 in the same position as the positive indices.

Direction		New Origin O'
$[\bar{1}11]$	→	(1,0,0)
$[1\bar{1}0]$	→	(0,1,0)
$[02\bar{1}]$	→	(0,0,1)



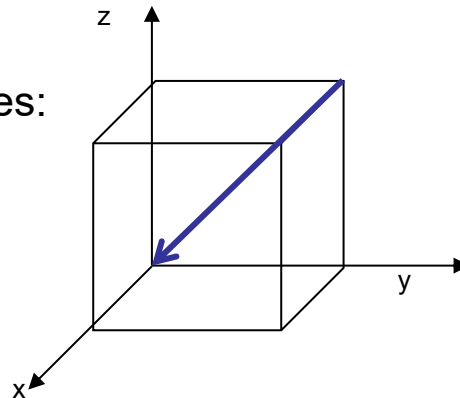
DIRECTIONS IN THE CUBIC CELL

Procedure for determining crystallographic directions:

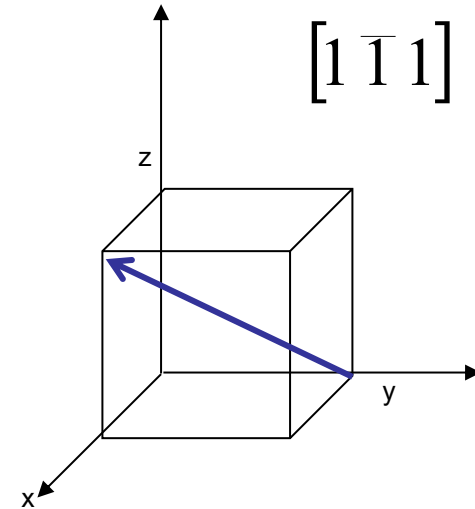
1. Choose a vector of appropriate length that passes **through the origin** (parallelism must be maintained).
2. Determine the length of the projection on each of the three axes *measured in terms of the unit cell dimensions a , b , c* .
3. Multiply or divide by common factor to reduce to the smallest integer value.
4. Notation: use $\bar{}$ for negatives and place square brackets, without comma [$y\ v\ w$]

Determine the indices:

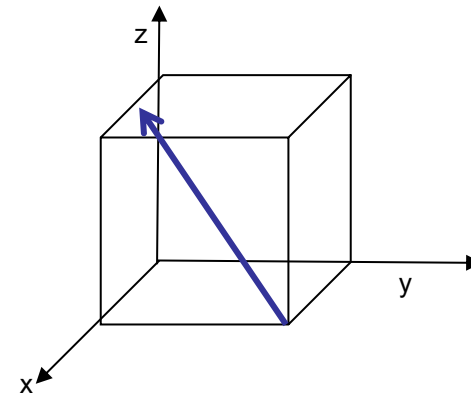
$$[0\ \bar{1}\ \bar{1}]$$



Examples



Draw the direction: $[\bar{1}\ \bar{2}\ 2]$



DIRECTIONS IN THE CUBIC CELLS

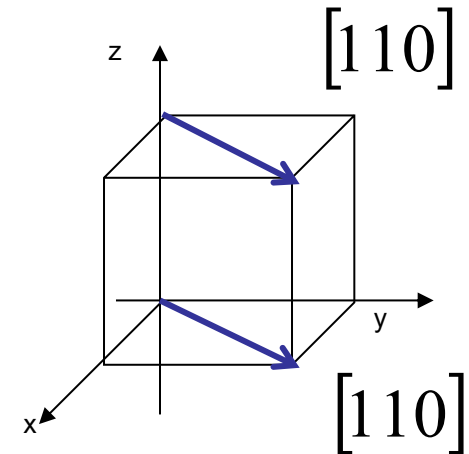
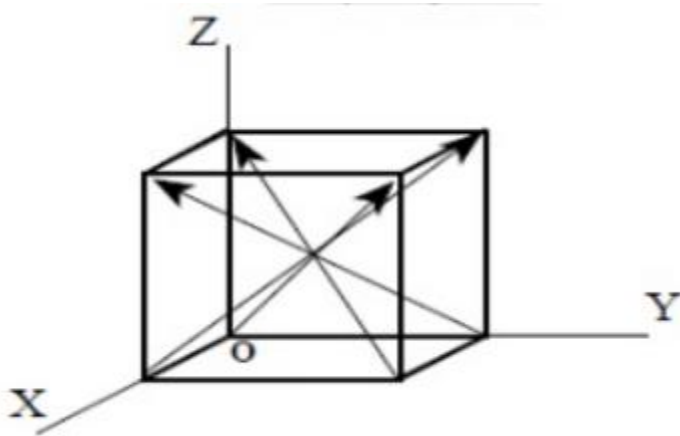
Family of directions

- Equivalent directions in the lattice form a family of directions (atom spacing along each direction is the same)
- **Notation:** $\langle u \ v \ w \rangle$

Cube axes: $[100], [010], [001], [0\bar{1}0], [00\bar{1}], [100] = \langle 100 \rangle$

Cube face diagonals: $\langle 110 \rangle$

Cube diagonals : $\langle 111 \rangle = [111], [\bar{1}\bar{1}\bar{1}], [\bar{1}11], [1\bar{1}\bar{1}]$



// direction vectors $\Rightarrow$ the same indexes

TOPIC 2. STRUCTURE OF MATERIALS I

- Crystalline systems and some definitions.
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- Directions in the crystalline cells.
- **Planes in the crystalline cells.**

PLANES IN THE CUBIC CELL

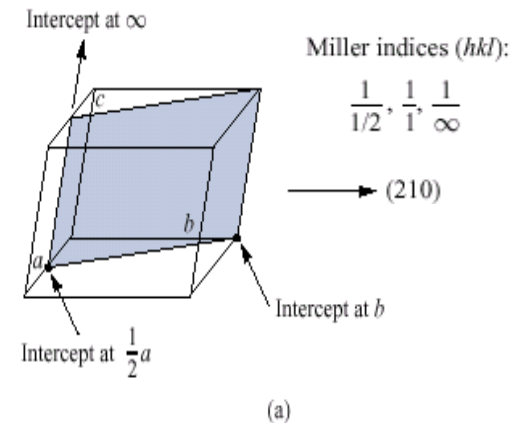
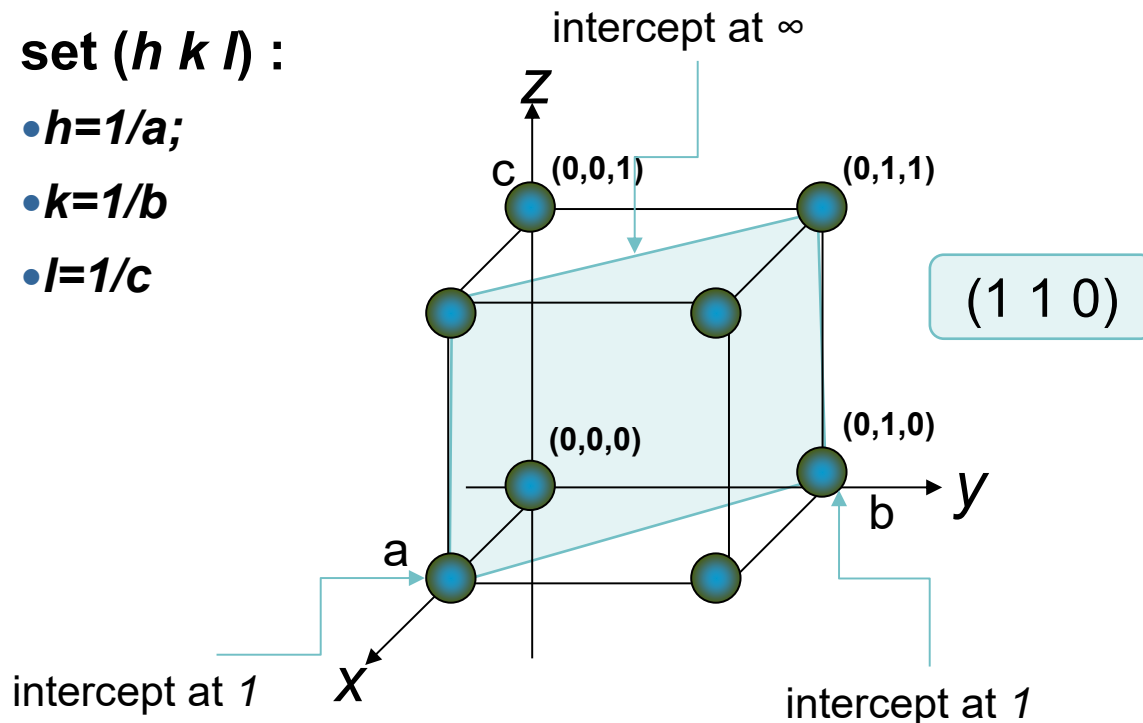
Crystallographic planes

Described by **Miller indices** ($h\ k\ l$)

- Miller indices ($h\ k\ l$): **reciprocal values** of the intersections of the crystallographic plane with a , b and c axes.

set ($h\ k\ l$) :

- $h=1/a$;
- $k=1/b$
- $l=1/c$

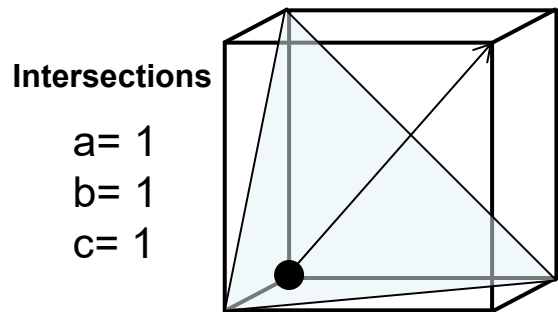


PLANES IN THE CUBIC CELL

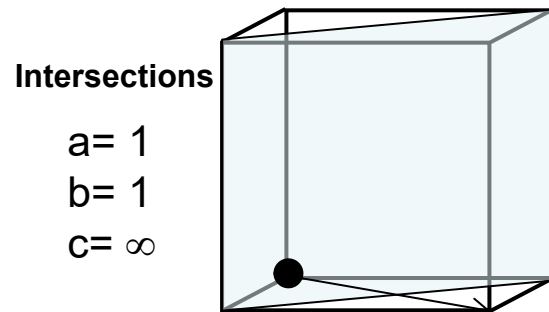
Crystallographic planes

Procedure for determination of Miller indices of a plane

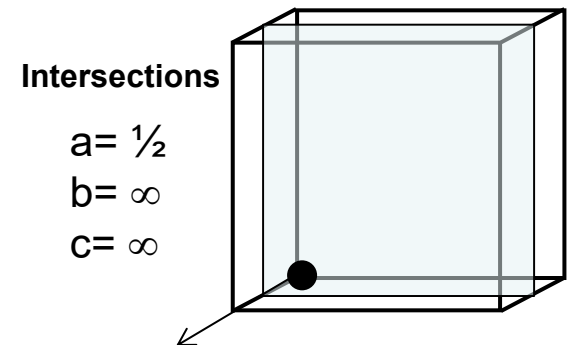
1. Choose origin O (the plane should not pass from the origin; choose the closest corner)
2. Determine the intercepts of the plane with the 3 axes
3. Find the **reciprocals** of these the intersections
4. Determine the smallest set of integers that have the same ratio as the intercepts
5. Notation: Enclose them in parentheses **without** coma ($h\ k\ l$)



Plane (111)



Plane (110)



Plane (200)

In **cubic** lattices, **directions** are $\perp$ to the **plane** with the **same Miller indices**

PLANES IN THE CUBIC CELL

Crystallographic planes

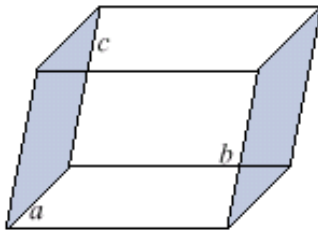
Examples

$$\left. \begin{array}{l} x = 1/2 \\ y = 1 \\ z = \infty \end{array} \right\} (210)$$

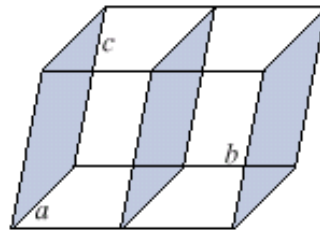
$$\left. \begin{array}{l} x = 1 \\ y = 1 \\ z = \infty \end{array} \right\} (110)$$

$$\left. \begin{array}{l} x = 1 \\ y = 1 \\ z = 1 \end{array} \right\} (111)$$

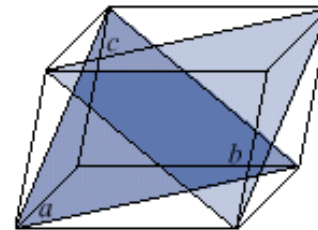
$$\left. \begin{array}{l} x = 1 \\ y = \infty \\ z = \infty \end{array} \right\} (100)$$



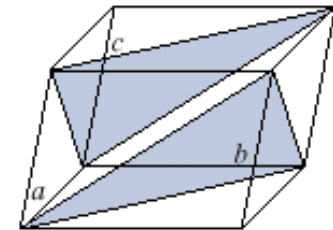
(010)



(020)

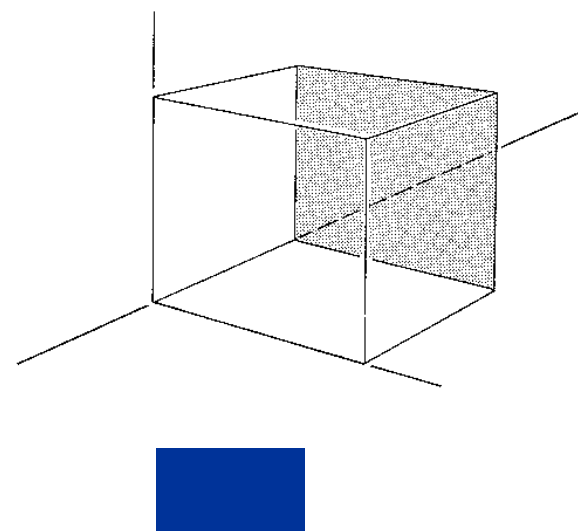
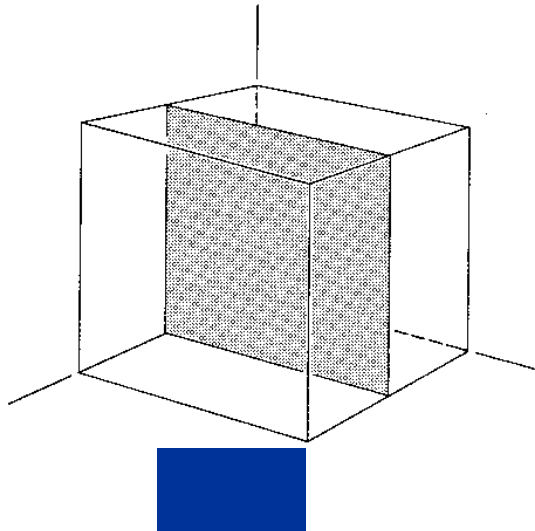
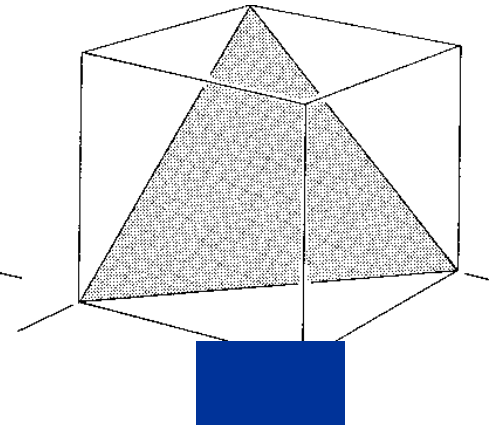
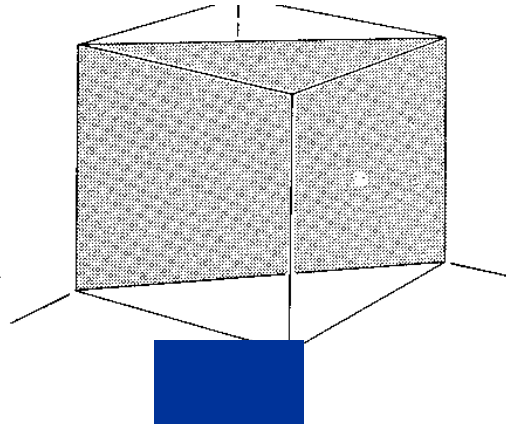
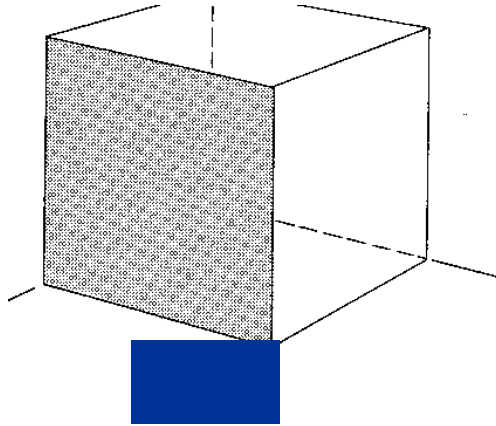


(111)



(111̄)

(b)

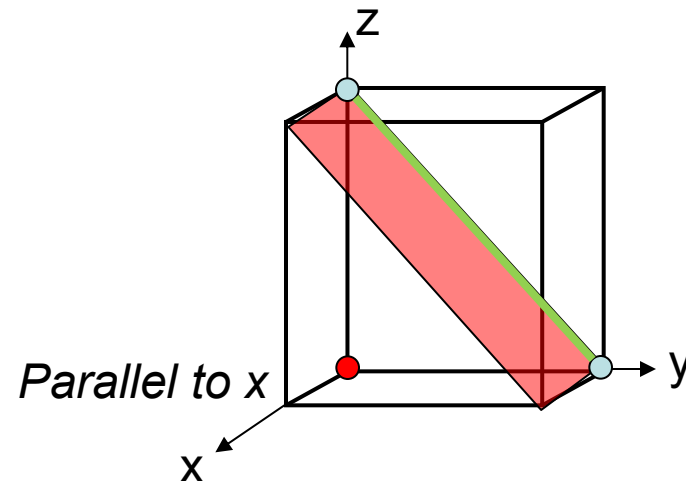
PLANES IN THE CUBIC CELL EXAMPLES**Examples**

PLANES IN THE CUBIC CELL

Procedure to draw a known plane (h k l)

Example: (0 1 1)

1. Select origin O at (0,0,0) (OR if there are negatives at O')
2. Take reciprocal of indices to get $1/h$, $1/k$, $1/l$: which will be the intercepts.
3. Mark intercepts along x, y, z at $a=1/0=\infty$; $b=1/1=1$; $c=1/1=1$
4. Draw the plane by connecting intercepts



PLANES IN THE CUBIC CELL

Procedure to draw a known plane (h k l)

Planes in the cubic cell with negative indices

If we want represent a plane in the positive axes: We have to move the origin of coordinate system in the negative index

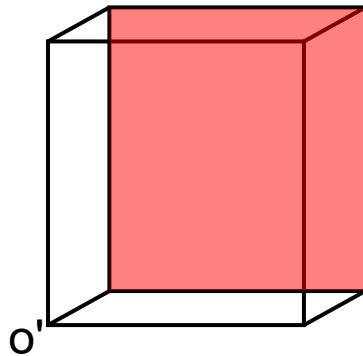
Plane ($\bar{1}00$)

New origin: (1,0,0)

$a = -1$

Intersections $b = \infty$

$c = \infty$



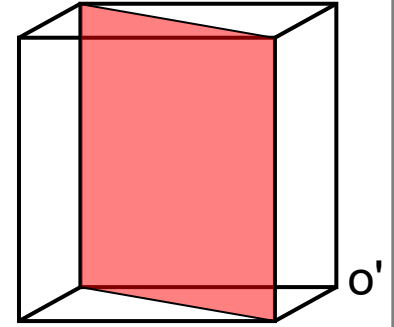
Plane ($1\bar{1}0$)

New origin: (0,1,0)

$a = 1$

Intersections $b = -1$

$c = \infty$



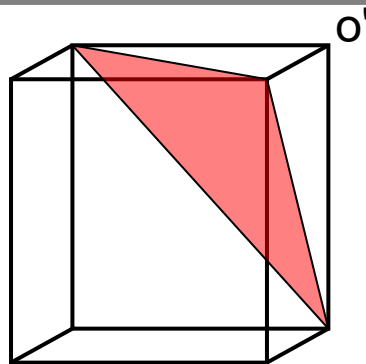
Plane ($1\bar{1}\bar{1}$)

New origin: (0,1,1)

$a = 1$

Intersections $b = -1$

$c = -1$



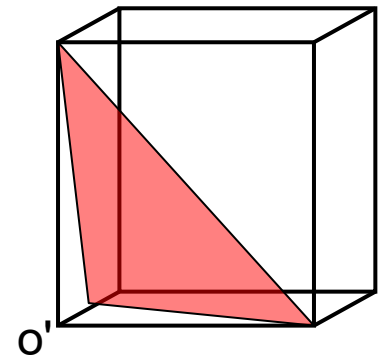
Plane ($\bar{2}11$)

New origin: (1,0,0)

$a = -\frac{1}{2}$

Intersections $b = 1$

$c = 1$



PLANES IN THE CUBIC CELL

Family of planes

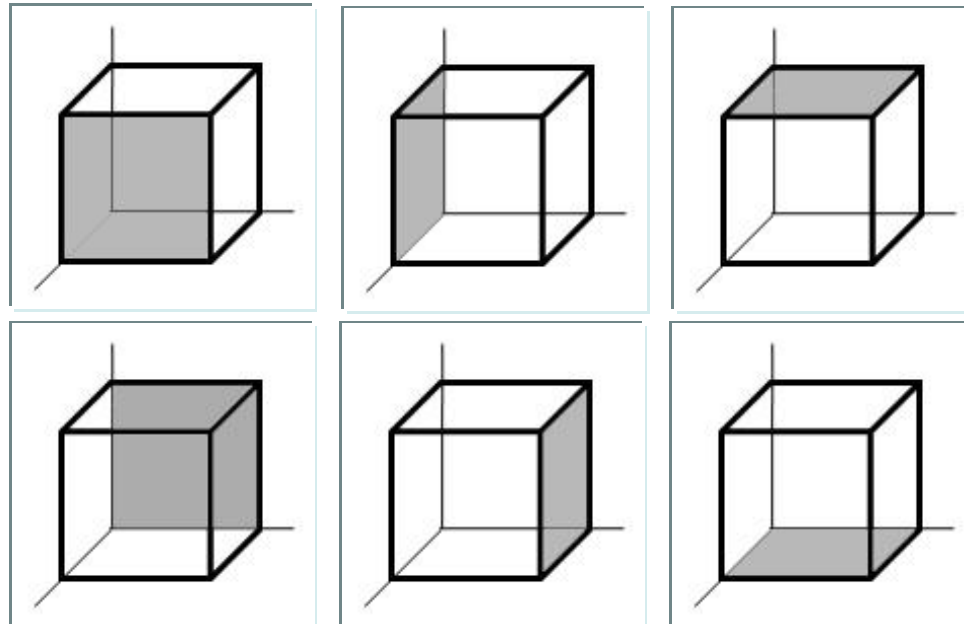
Planes || equidistant $\Rightarrow$ Same Miller indices

Planes of a family: $\{H K L\}$ Two planes belong to the same family if they have similar atomic distributions.

$$(100)(010)(001)(\bar{1}00)(0\bar{1}0)(00\bar{1}) = \{100\}$$

$$(111)(1\bar{1}\bar{1})(\bar{1}11)(\bar{1}\bar{1}1)(\bar{1}\bar{1}\bar{1})(\bar{1}1\bar{1})(1\bar{1}1)(11\bar{1}) = \{111\}$$

Example:
Family $\{100\}$
Cube faces



DIRECTIONS AND PLANES IN THE HEXAGONAL CELL

In this case **there are 4 axes**: 3 in the basal plane and the c axis

Basal plane (a_1, a_2, a_3) + Axis c

$\downarrow$ $\downarrow$
 h, k, i l

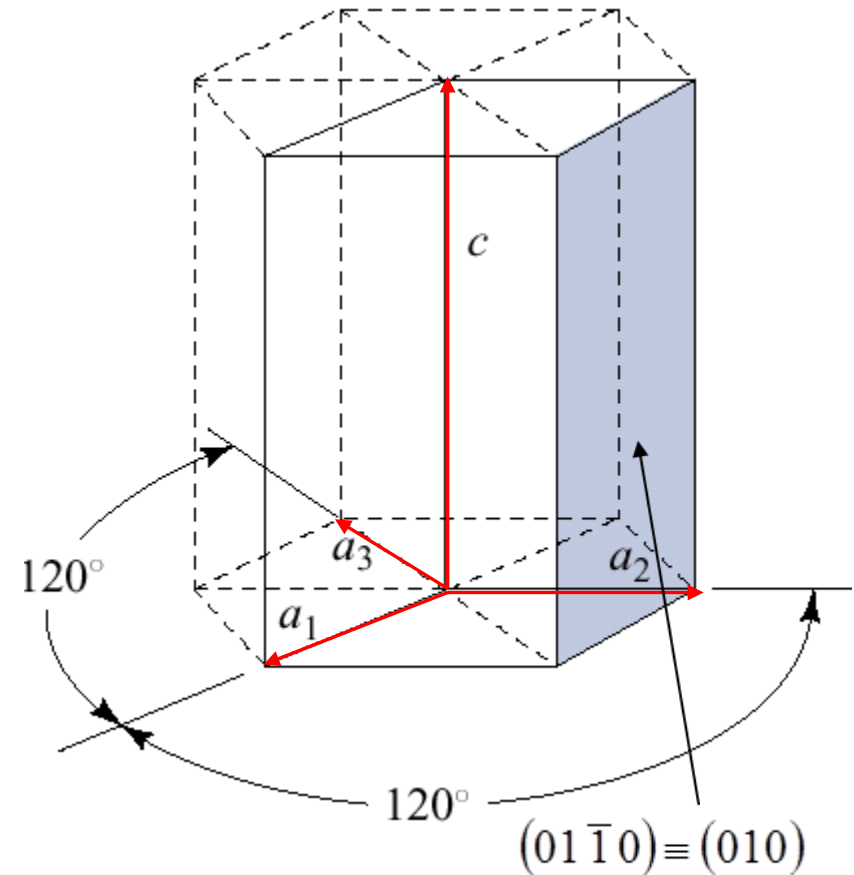
- Planes**

They can be expressed in 3 taking

$$i = -(h+k)$$

$$(h' k' l') \equiv (h k i l)$$

$$(0 1 0) \equiv (0 1 \bar{1} 0)$$



DIRECTIONS IN THE HEXAGONAL CELL

• Directions

To convert a direction from 3-index to 4 index:

$$[h' \ k' \ l'] \equiv [h \ k \ i \ l]$$

$$h = \frac{1}{3}(2h' - k')$$

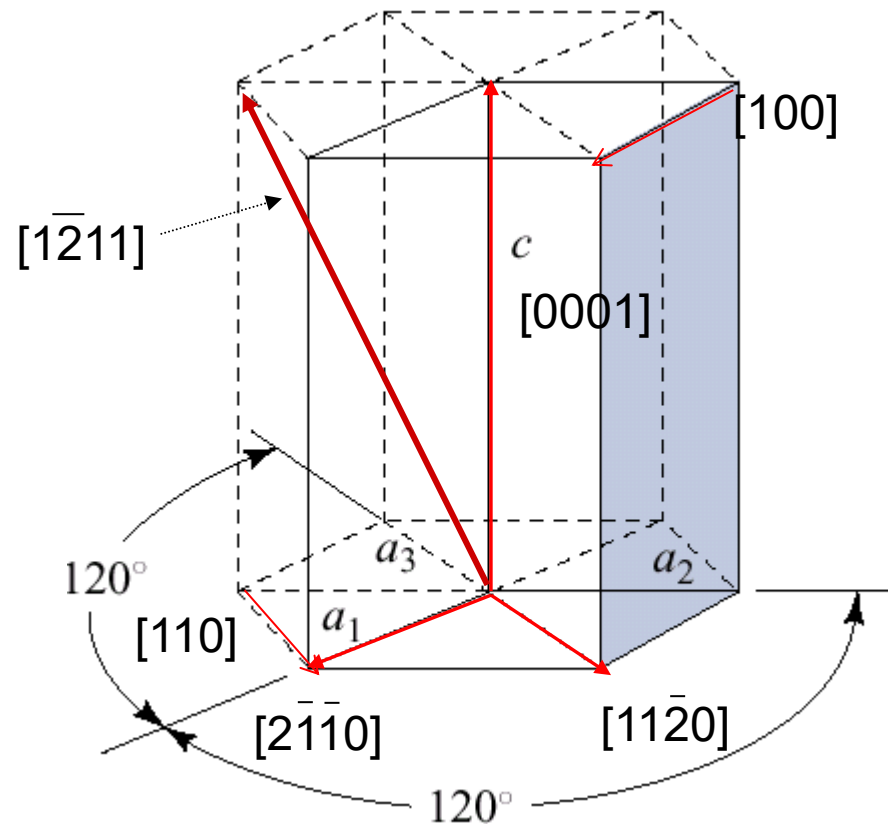
$$k = \frac{1}{3}(2k' - h')$$

$$i = -\frac{1}{3}(h' + k')$$

$$l = -l'$$

$$[1 \ 0 \ 0] \equiv [2 \ \bar{1} \ \bar{1} \ 0]$$

$$[1 \ 1 \ 0] \equiv [1 \ 1 \ \bar{2} \ 0]$$



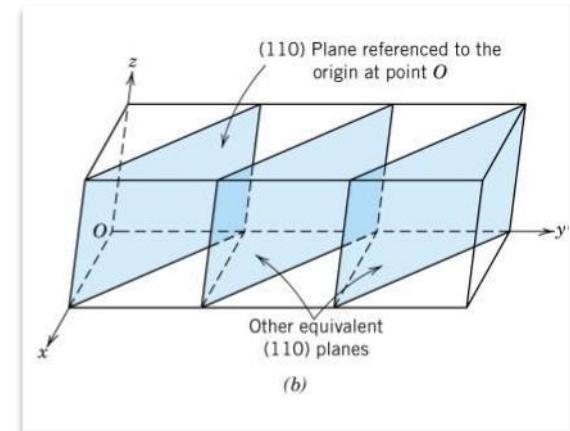
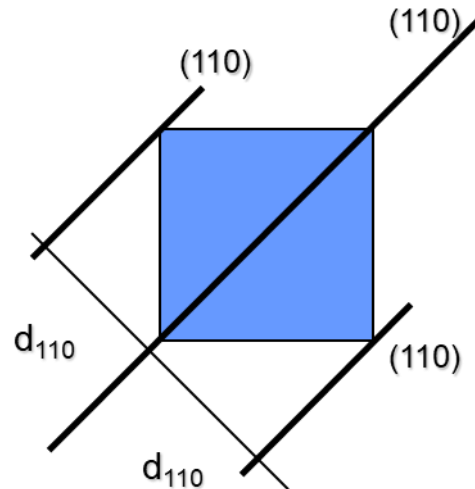
Bragg's law

The distance between two adjacent parallel planes (the **interplanar spacing** d_{hkl}) is a function of the Miller indices (h, k, l) and the lattice parameter(s)

Cubic Crystals:

- Distance between // planes in cubic crystals : d_{hkl}

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$



- For **orthogonal** crystals, in general:

$$\frac{1}{d_{hkl}^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$